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HS1 Radio Astronomy

(public version)

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KRT 3 radio telescope at Hamburg Observatory

Note: This is the public version of this document, which contains slightly less information. If required get in touch with the telescope supervisor (Volker Heesen) to obtain the internal version intended for use at the Observatory.

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1 Background

Modern astrophysics requires the use of the entire electromagnetic spectral range for the comprehensive investigation of the processes in the Universe and their interactions. Besides optical radiation, only radio waves with wavelengths in the centimetre and metre range reach the Earth's surface unhindered. Research into the universe with radio waves began after the Second World War and is now an established branch of astrophysics.

In Germany, the 100-metre Effelsberg radio telescope located in the Eifel was inaugurated in 1971. It was the world's largest fully mobile single antenna for a long time with a large number of receivers available for the radio frequency range.¹ Since 2007, several German institutes have also been involved in the world's largest radio interferometer, the LOw Frequency ARray (LOFAR), with 6 observation stations.² The last German station was built in Norderstedt near Hamburg and was inaugurated in September 2015. It was set up by the Universities of Hamburg and Bielefeld, thus giving their research groups privileged access to the low-frequency spectral range from 30 to 240 MHz.

Radio astronomy has several advantages over other parts of the electromagnetic spectrum. For example, radio waves are not absorbed by dust in galaxies unlike optical waves. Radio astronomy is also the only ground-based astronomy, apart from the optical branch, which of course makes the operation of telescopes much easier. In addition, radio astronomy has the advantage that interferometry is relatively simple and can thus produce spatial resolutions at least equal to optical astronomy, despite the long wavelengths. For instance, the LOFAR radio interferometer can achieve an angular resolution of only 0.3 arc seconds despite a wavelength of 2 metres!

The Hamburg Observatory operates a small radio telescope (KRT 3) primarily for teaching purposes. It is a single-dish telescope with a diameter of 3.0 metres and equipped with a receiver designed to receive the 21-cm line radiation (about 1420.4 MHz) of neutral interstellar hydrogen (H I). This line is produced by a hyperfine structure transition during a so-called *spin flip* of the hydrogen atom. Since it is line emission, the Doppler shift can be used to determine the radial velocity. With the help of the instrument, the basic radio astronomical observation methods with single-dish telescopes can be reproduced. The measurement of the 21-cm line along the Galactic plane allows us to measure the velocities of the gas at different points in the disc of the Milky Way. Thus, both the structure of the spiral arms and the rotational direction of the Milky Way can be determined (see Fig. 1).

The aim of the experiment is to determine the *rotation curve* of the Milky Way within the orbit of the Sun. With its help, a rough estimate of the mass of the Galaxy can be made. It can be seen that the Milky Way actually rotates too fast in the peripheral regions, if only the stellar mass is taken into account. This observation, and especially that of other nearby galaxies, was the first indication in astronomy of the mysterious dark matter, the nature of which is still unknown today and is the subject of current research. Our galaxy, for example, has a stellar mass of about 10^{11} solar masses and a total mass of about 10^{12} solar masses. So you can see that the dark matter here represents about the 90% of the mass of the Milky Way.

¹see: <http://www.mpifr-bonn.mpg.de/effelsberg>

²see: <https://www.glowconsortium.de/index.php/en/lofar-about>

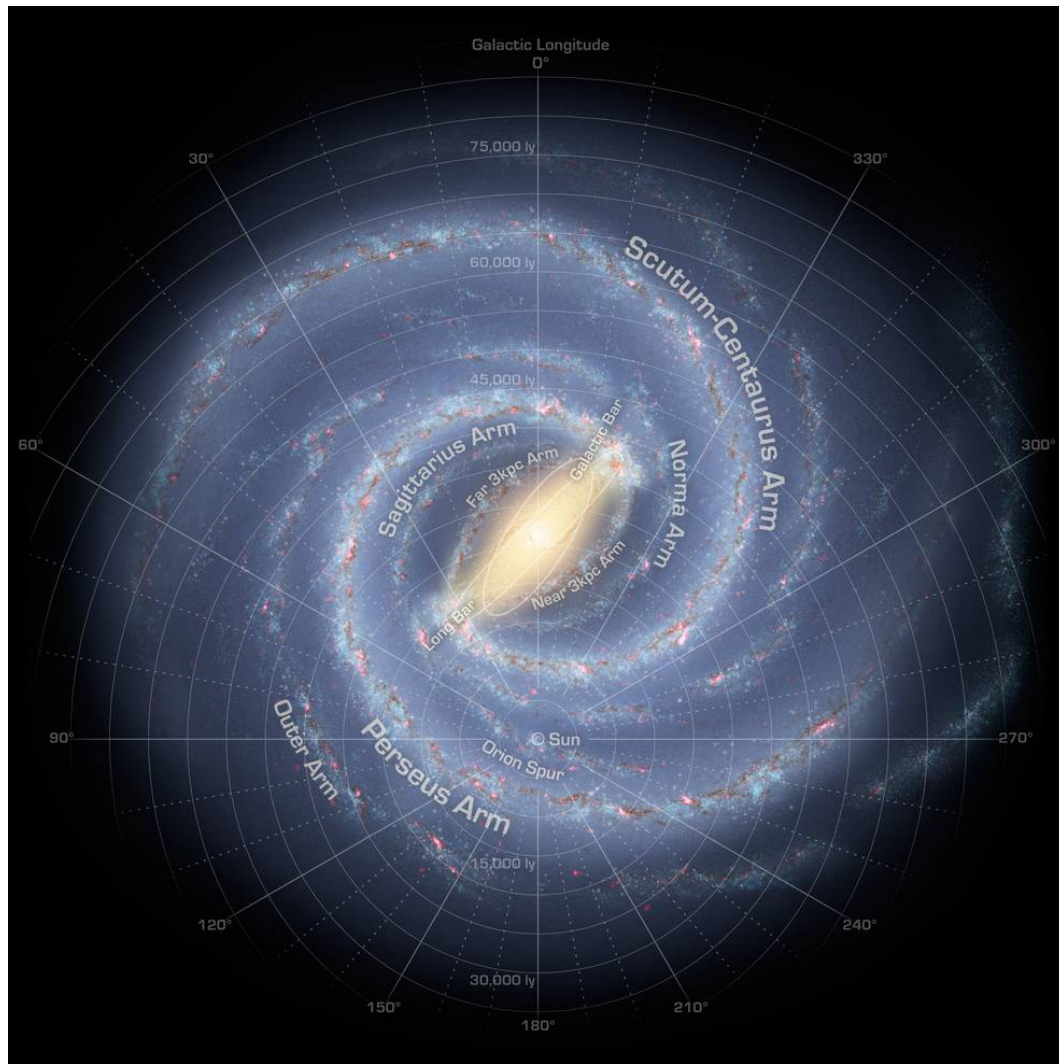


Figure 1: *Artistic impression of the Milky Way.* This view of the Milky Way is inspired by observations and represents the current state of research. Source: NASA/JPL-Caltech/R. Hurt (SSC/Caltech)

This experiment instructions (Versuchsmappe) for the Advanced Lab Course (F-Praktikum) experiment ‘HS1 Radio Astronomy’ consists of several parts. In Section 2, the necessary pre-requisites are discussed. Section 3 provides an overview of the KRT 3 radio telescope. Section 4 explains how to operate the KRT 3. The main part of the manual contains Section 5, where the individual experiments are explained. This includes the calibration of the telescope (Section 5.1), the determination of the spatial resolution (Section 5.2), the determination of instrument’s sensitivity (Section 5.3) and finally the observation of neutral hydrogen in the Milky Way (Section 5.4). Section 6 explains what needs to be considered at the end of the observations. Section 7 gives some hints on the evaluation of the data. Section 8 presents a possible structure of the experimental protocol. Finally, there is an appendix with hints on how to solve problems during KRT 3 operations (Appendix A), some command line commands for LINUX (Appendix B), a mini-introduction to PYTHON (Appendix C) and a short history of the KRT 3 (Appendix D).

2 Requirements for conducting the experiment

2.1 Necessary prior knowledge

In order to be able to carry out the experiment successfully, you may want to familiarise yourself not only with this experiment map but also with the following documents, which you can have access to by request.

- Manuscript *HSI Radio Astronomy – Background*. It introduces the concepts and terms used in radio astronomy;
- Instructions *SPIDER Radio telescope user manual*. It describes the operation of the radio telescope in detail.

In particular, one should read the *Background* in order to have the necessary prior theoretical knowledge. This prior knowledge will then be tested during the preliminary discussion. Also, ideally familiarise yourself with Section 4 of this manuscript and do the preparatory work described in Section 4.1 before you start the observations.

2.2 Computer equipment

If desired, the analysis can be carried out entirely on a computer at the observatory. However, it is helpful to bring your own portable computer, which should ideally be a *laptop* with keyboard although a tablet computer with an *external keyboard* will be also fine. Since smaller programming tasks are to be carried out, a screen keyboard is not ideal. A *computer mouse* has also proven to be helpful in order to be able to examine the H I-spectra in detail.

The *Nextcloud* also contains the sub-folder *python*. There you will find two PYTHON scripts. Ideally, before starting the experiment, you should check that both scripts are running on the computer you have chosen for the data analysis. To do this, open a terminal and enter the following commands:

```
python fit.py buch.txt
```

A two-dimensional diagram should then open that can be modified interactively. If this works so far, also execute the next command:

```
python spectrum.py l110.fits
```

This allows you to display an H I spectrum with archival data. If you have difficulties with this, you will find some hints in Section C on how to do it. In particular, an up-to-date PYTHON-installation is recommended with the ASTROPY-package with a version number of $v \geq 4.0$.



Figure 2: *KRT3 radio telescope*. Parabolic antenna with 3.0 m diameter and receiver part in primary focus (top left). Alt-az *WP-100* mount with control unit (top right). Part of the dish during assembly in summer 2019 with a mobile crane (bottom left). Photograph taken with a drone above the roof of the library (bottom right).

3 The KRT 3 radio telescope

In simple terms, a radio telescope works in a similar way to a radio receiver as known from radio communication. It is an electronic device that receives radio waves and converts the signals they transmit into a usable form. It is used with an antenna consisting of the parabolic dish and the receiving horn. The antenna intercepts radio waves (electromagnetic waves) and converts them into tiny alternating currents that are applied to the receiver, and the receiver extracts the desired information. The receiver uses digital filters to separate the desired frequency from all other frequencies picked up by the antenna. The pre-amplifier is crucial to increase the power of the signal for further processing. Finally, the digitised signals are further processed with software. The KRT 3 used in the experiment is presented in more detail below.

The KRT 3 is a SPIDER 300A-model from the company PrimaluceLab SpA from Italy, which was installed in summer 2019 (see Fig. 2). It is a radio telescope with a parabolic mirror of 3.0 m diameter and a digital receiver. The *WP-100* mount of the telescope is an *alt-azimuth mount*, i.e.



Figure 3: *Control and receiver of the KRT3 radio telescope. Left: Server cabinet with the RCPU-100 mount control unit, the H142-One receiver and the Spider-PC for control and data reduction. Right: Receiving horn in the primary focus of the antenna. The so-called septum splits the electromagnetic radio waves into left-hand circular polarisation (LHCP) and right-hand circular polarisation (RHCP).*

the antenna is freely movable in both elevation and azimuth and can be automatically tracked in azimuth and elevation. The spatial resolution capacity of the antenna is small and amounts to a multiple of the angular diameter of the Moon. A thin wire grid mounted onto an aluminium frame reflects the incident radio radiation and focuses the radiation onto a receiving horn.

The technical properties in the overview are:

- Receiver type: digital superheterodyne receiver;
- Frequency range of the preamplifier: 1398–1460.5 MHz;
- Half-width of the antenna lobe: about 5° ;
- Positioning accuracy: 1° .

The circular receiving horn *H-FEED* is located at the primary focus of the antenna. It is an electrically conductive circular horn in which the radio waves are split into left-circular and right-circular polarised waves by means of a so-called septum (Fig. 3). The receiving horn is constructed in such a way that lateral scattered radiation (*spillover*) is reduced as much as

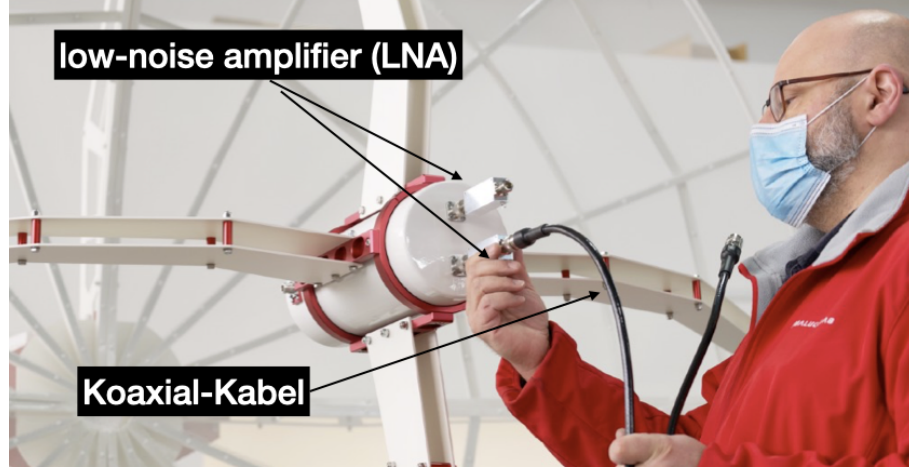


Figure 4: *Low noise preamplifier*. The low-noise preamplifiers (LNAs) for the left and right circular polarisations are installed directly behind the horn to minimise signal loss. Source: PrimaLuceLab SpA.

Table 1: Default setting of the KRT 3 receiver The individual frequency bands are referred to as *Base Band Converters* (BBC) and are available for both left (L) and right (R) circularly polarised signals.

BBC	Pol.	Channels	Frequencies (MHz)	BBC	Pol.	Channels	Frequencies (MHz)
1	L	0001–0128	1398.0000–1405.8125	09	R	0001–0128	1398.0000–1405.8125
2	L	0129–0256	1405.8125–1413.6250	10	R	0129–0256	1405.8125–1413.6250
3	L	0257–0384	1413.6250–1421.4375	11	R	0257–0384	1413.6250–1421.4375
4	L	0385–0512	1421.4375–1429.2500	12	R	0385–0512	1421.4375–1429.2500
5	L	0513–0640	1429.2500–1437.0625	13	R	0513–0640	1429.2500–1437.0625
6	L	0641–0768	1437.0625–1444.8750	14	R	0641–0768	1437.0625–1444.8750
7	L	0769–0896	1444.8750–1452.6875	15	R	0769–0896	1444.8750–1452.6875
8	L	0897–1024	1452.6875–1460.5000	16	R	0897–1024	1452.6875–1460.5000

possible. Behind the horn are two low-noise pre-amplifiers, one for each polarisation, which are protected from environmental influences by a white plastic housing (Fig. 4).

The radiation reflected by the antenna is thus first amplified by a factor of 33 dB by the low-noise amplifier (LNA) and then passes through a bandpass filter. The arrangement of the LNA directly behind the horn minimises the signal losses before the first amplification stage. The preamplifier has a frequency range of 1398–1460.5 MHz (λ 20.5–21.4 cm; band around the 21-cm line) with a bandwidth of 62.5 MHz. The effective frequency is therefore 1429.25 MHz, which corresponds to an effective wavelength of $\lambda_{\text{eff}} = 21.0$ cm. The signal is then fed through the mount using two coaxial cables and routed to the *H142-One* receiver (Fig. 5).

The bandwidth is divided by the receiver into 1024 channels, each with a bandwidth of 61 kHz. The bandwidth is further subdivided into 16 frequency bands with the so-called *base band converters* (BBC) (see Table 1). After amplification, the signal is superimposed with the signal of a local oscillator, whereby it is down-modulated to a lower intermediate frequency. The signal is then digitised with the aid of a 14-bit analogue-to-digital converter. The spectrum is

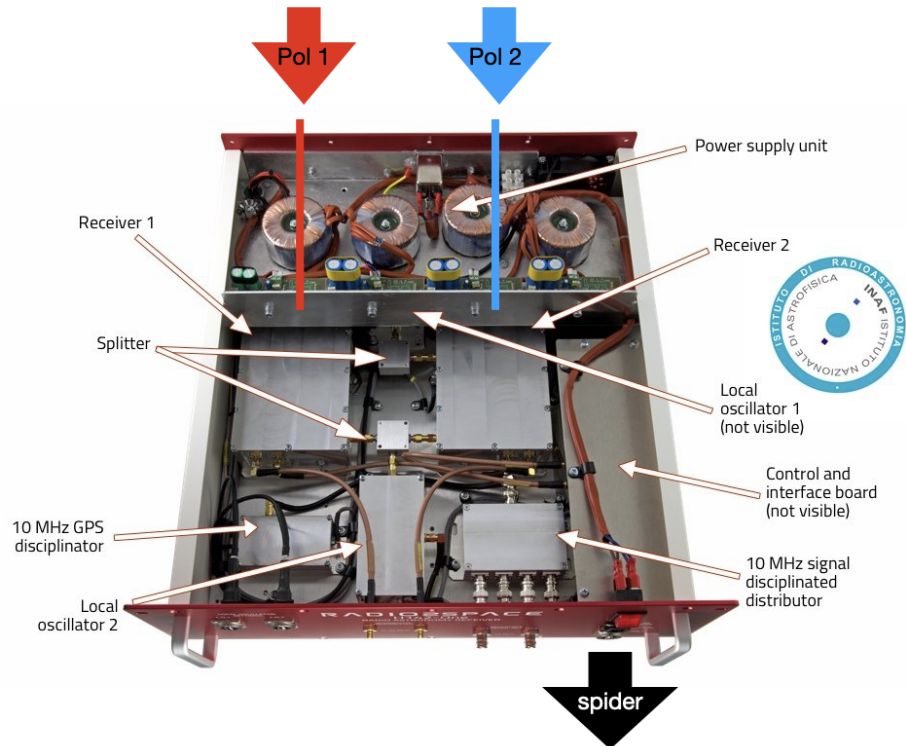


Figure 5: *H142-One* receiver. The incoming signal of the two circularly polarised waves is mixed down with the help of two local oscillators (LO). A 14-bit analogue-to-digital converter (ADC) is used to digitise the signals. Source: PrimaLuceLab SpA

calculated from the digitised signal with a ‘digital signal processor’ using a Fourier transform. The telescope mount is controlled by the *RCPU-100* unit, which is connected to the *spider* control PC via a USB interface (see Fig. 3). Analogously, the *H142-One* is then connected to the control PC via an *Ethernet* cable.

The software RADIOUNIVERSEPRO allows one to control the telescope, record measurement data as well as display the data. It runs on the dedicated PC *spider* to which one can log in from all computers of the Observatory. A virtual desktop is used for this purpose, and the programme REMMINA is applied here. For monitoring the KRT 3 a webcam is available. This allows one to conduct the radio observations from any desktop computer at the Observatory. During the lab course, the computer *pool02* is used, which is located in *room 109* (meeting room, Besprechungszimmer) on the first floor of the main building (library, Bibliothek).

4 Operating the KRT 3 radio telescope

At the beginning of the lab course, you will receive an introduction from your supervisor how to operate the KRT 3. Detailed operation instructions can be also found in the *SPIDER radio telescope user manual*. Read the description in advance so that you can quickly find the nec-

essary information in case of operating questions during the experiment. Keep a measurement log (see Section 4.3). Note down any steps that deviate from these instructions and write a short weather report (temperature, wind conditions, precipitation).

4.1 Preliminary work

As a ‘Quick Start guide’, you may perform the following steps in order to start operating the telescope:

- Log in to the lab course desktop computer. This computer works with a LINUX operating system;
- Open a window by starting the terminal programme (hereafter: working window) to be able to enter commands on the command line. The most common commands for the command line can be found in Appendix B;
- You are in the home area of the desktop computer. Enter `pwd` and you will get the address of the home area:

```
prak@pool02:~$ pwd
```

```
/export/data/pool02/praktikum
```

Change there to the *Desktop* working directory where you can store your data.
Command: `cd Desktop`;
- Change to the directory. Command: `cd [directory_name]`. A renewed `pwd` gives as answer `cd [directory_name]`.

```
prak@pool02: /data/pool02/praktikum/[directory_name];
```
- You can display the contents of your working directory with the command `ls`. It should contain at least the file *spectrum.py*.
- Your measurement log should now at least list the names and contents of the directories and files you have created on the lab computer.

4.1.1 Logging in to the radio telescope server *spider*

The telescope software runs on a dedicated PC, namely the telescope control server *spider*. The full address is `spider.hs.uni-hamburg.de`. It is located in the attic of the main service building (library). The operating system is *Windows10*. The *spider* PC can be accessed from any desktop computer at the Observatory with the remote desktop client REMMINA. Alternatively, *spider* can be accessed from outside the Observatory if a *Virtual Private Network* (VPN) connection with the Hamburg University network has been established. The Cisco VPN client



Figure 6: *Telescope control panels*. The panels are divided into three sections. On the left is the telescope control, on the right the receiver setting. In the middle (in black) are displays that can be controlled with the button bar. There are also displays for the command line and log messages (from top to bottom).



Figure 7: Telescope control panel display after connecting to the mount (left) and receiver (right).

from the Regional Computer Centre (RRZ) of the University of Hamburg is available for all current systems ³.

Note that it is advantageous to share a directory so that it is relatively easy to copy data from *spider* to the local PC. The screen resolution of the local PC should be at least Full HD (1920 × 1080 Pixel²) so that the control software of the telescope can be fully displayed.

You can switch to the local PC's workspace at any time using the REMMINA control panels at the top of the screen (it hides there behind the shadow!). The best way to work is in full screen mode.

³<https://www.rrz.uni-hamburg.de/services/netz/vpn.html>



Figure 8: *IF Monitor*. The windows display the following information: Status of the mount (1), orientation of the telescope in azimuth (2), status of the receiver (3), status messages in text form (4), spectra with left and right circular polarisation.

4.1.2 Quick start

After logging on to *spider*, the telescope control software RADIOUNIVERSEPRO can be started on the remote desktop and the telescope control panel (Fig. 6) appears. First connect to the mount (Fig. 7, left). In the ‘Mount Control’ section, select the *mount type* WP-100 (default) and press *Connect*. The displays with the coordinates will then fill up. The telescope is initially in the **park position** at Azimuth = 270° and Elevation = 90°. These values are displayed in the *Actual Antenna Coordinates* subsection. Then connect to the receiver (Fig. 7). In the *Receiver and Data Acquisition Control* section, click *Connect* to backend and then later when ready *Start Aquisitions*.

You can point the telescope using the command line in RADIOUNIVERSEPRO using the command:

```
source=azel,270,85
```

This will point to KRT3 to the azimuth of 270° and an elevation of 85°. These values can be adjusted as needed, of course.

The telescope is now ready for observations! ☺

4.2 First observations

4.2.1 Starting the telescope

In order to start up the telescope, perform the following steps:

- Open another tab (Strg+Shift+T) (hereafter: *observation window*) and change to your working directory (Desktop);
- Log in with a virtual desktop on the PC *spider*. To do this, execute the command `remmina`; in the application that is now open, double-click on *spider* to start the connection. After that, you see the desktop on the spider-pc. Double-click on the RADIOUNIVERSE-PRO software to start the control software (Fig. 8);
- First select the *WP-100* mount (*control panel 1*) and click on Connect. In the control panel, the red line should now be aligned to the west;
- In *control panel 3*, click on Connect to Backend. In log window 4 you will see the output. When the receiver is ready, click on Start Aquisitions (button 3);
- Now the two spectra (left and right circular polarised) should be visible as green lines in window 5.

4.2.2 Adjusting the receiver settings

Next, we adjust the receiver settings for radio continuum observations. For this we use the Base Band Configuration Tools (*BBC Tools*):

- Now switch to the *BBC Tools* (Fig. 9);
- The two windows show the spectra in the respective left and right circular polarisation. The y-axis shows the un-calibrated antenna temperature in units of counts per second. The channels are shown on the x-axis. There are 1024 channels covering the frequency range from 1398 to 1460.5 MHz.
- In *BBC Setup* (window 2) select Upper Side Bands (USB), i.e. from BBC to 01 BBC The 16. The colours of the boxes indicates how good the quality ($QF = 0-1$) of a frequency band is;
- Confirm the selection of the frequency bands with Save new setup (Window 2);
- Now adjust the gain so that the spectra (window 1) have a Total Power (TP) at the maximum of about 20.000. To do this, use the IF Gain in window 3;
- Now the intensity histograms in windows 4 should also become clearly visible.

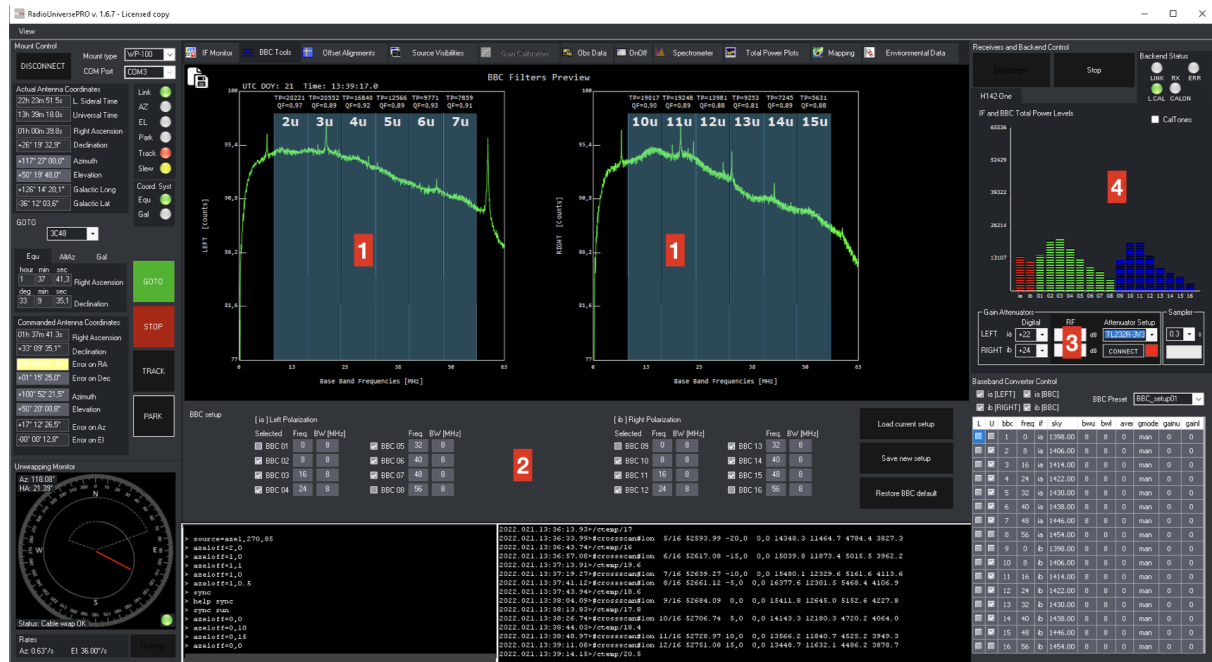


Figure 9: *BBC Tools*. The windows display the following information: uncalibrated spectra of left and right circular polarisation (1), selection of frequency bands (2), receiver gain setting (3), bar graphs of the measured total power intensity in the frequency bands (4).

4.2.3 Checking the positioning accuracy

Next, we observe the Sun to check the positioning accuracy of the KRT 3 (see Fig. 10). For this purpose, the following steps are carried out:

- Point the telescope at the Sun by going to the *Source Visibilities* tab. There, double-click on *Sun*. Wait until the TRACK button changes to STOP TRACK;
- adjust the gain as described in section 4.2.2;
- then go to the *Offset Alignments* tab;
- in the *Pointing Settings* window, make the settings as shown in Fig. 10 (left). Make sure that the *Step size (HPBW frac)* is set to 1;
- now click on Confirm setup;
- finally click on Start procedure.

When the KRT 3 has completed the procedure, the *Alignment Procedure* window should show a plot as in Fig. 10 (right). The crosses represent un-calibrated in-situ measurements, shown once as a function of horizontal deviation (Longitude Offset; left) and as a function of vertical deviation (Latitude Offset; right). If the positioning accuracy is good, a well-defined maximum should be seen at an offset of both 0° longitude and latitude.

Pointing Settings			
Repetitions	1	<	>
Integration period [s]	2	<	>
Approx. duration [s]	28		
Points per axis	7	<	>
Step size (HPBW/frac)	1	<	>
Field of pointing [deg]	28,24		
Antenna offsets			
Commanded	Actual	Commanded	Actual
Az:	Et:	Az:	Et:

longitude

latitude

Confirm setup

Start procedure

Save as defaults

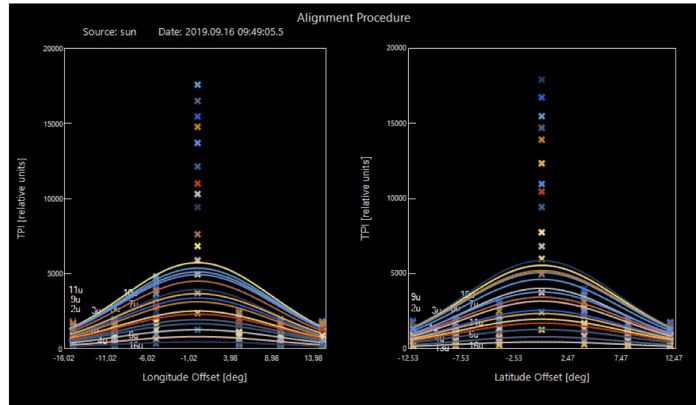


Figure 10: *Offset alignment*. In the left panel, the selected settings for correcting the positioning accuracy with the help of the Sun are shown. After successful correction, the diagrams look as shown in the right panel.

4.2.4 Pointing correction

If the positioning accuracy is not very good, then the encoders of the assembly should be reprogrammed. This is generally known as *pointing correction*.

- Select the tab *BBC Tools*.
- Note the maximum TP value in one of the frequency bands (e.g. BBC 2u);
- This value should be approximately 20.000. If this is not the case, select a different gain as described in Section 4.2.2;
- Move 2° in azimuth direction away from the Sun using the command `azeloff=2,0` in the command line of RADIOUNIVERSEPRO (grey line left below). First you have to click exactly on the grey line, otherwise it is not active;
- Watch the TP value. If it rises, the KRT 3 is closer to the Sun. If the TP value decreases, move in the other direction with `azeloff=-2,0`;
- Find the ideal azimuth offset. An accuracy of $\pm 1^\circ$ is sufficient. In the following we assume that a correction of 4° was ideal, i.e. `azeloff=4,0`;
- Then repeat the procedure for the deviation in elevation. So start with `azeloff=4,2`. If the TP value increases, search in this direction (for positive elevation corrections). Otherwise, search in the other direction for negative corrections, i.e. `azeloff=4,-2`.

The ideal pointing correction is found when the TP value becomes maximum. Then adopt this value, e.g. with `azeloff=4,-4` for a correction of $+4^\circ$ in azimuth and -4° in elevation. Now the encoders must be reprogrammed so that the pointing correction is permanently adopted. To do this, type in the RADIOUNIVERSEPRO command line:

sync sun

The KRT 3 is now ready for use.

4.3 Measurement log

It has proved useful to keep a measurement log. The observations are entered in chronological order with the corresponding general settings and the receiver settings. The former includes the object names together with the values of azimuth and elevation angles. The latter includes, for example, the selected frequency bands (*BBCs*) and gain factors (*Gain*). In particular, it should be noted if there is any deviation from the instructions in the experiment folder. At the beginning of an observation, the date and the weather conditions should of course be noted.⁴ During the experiment, problems should also be noted, such as interfering radio frequency interference (RFI), which could falsify the measurements (see also Section 7.2). Pointing corrections made (Section 4.2.4) should also be noted here.

5 Experiments

5.1 Calibration

For each measurement, the receiver receives radiation from the source described by T_s as well as radiation from the environment described by T_h . In addition, there is also the radiation of the receiving system including the receiver and amplifier, which can be described by the system temperature T_{sys} . The measured power of the receiver represented by the antenna temperature T_A thus results in:

$$T_A = T_s + T_h + T_{\text{sys}}, \quad (1)$$

where T_{sys} is the pure system temperature. For simplicity, it is assumed that the source only covers a small part of the sky and thus, does not reduce the background radiation. Since it is difficult to separate the influence of the pure system temperature and the background radiation, these two temperatures are usually combined to form an effective system temperature:

$$T'_{\text{sys}} = T_{\text{sys}} + T_h, \quad (2)$$

so that T_h only contains the antenna independent contributions. The effective system temperature thus includes all (unwanted) contributions that do not come from the source. In order to determine the contribution of the source T_s at a measured antenna temperature T_A , one must know the effective system temperature. Determining this is the aim of calibrating a radio telescope.

⁴A suitable weather station can be found here: <https://wetter.ls.haw-hamburg.de/tagesarchiv.html>

In the following, we no longer distinguish between pure and effective system temperature, but only use the term system temperature, which stands for T'_{sys} .

To determine the system temperature, one uses a so-called ‘On–Off’ measurement by measuring the antenna temperature once with a source of known flux density and once at an empty position in the sky. Since one can assume in the first approximation that the radiation from the environment T_h is constant, the system temperature is then also known for measurements at other sky positions.⁵ The background temperature T_h is however elevation dependent, e.g. the sky background is higher at low elevations or there may be vegetation in the field of view of the antenna. For higher accuracy, this dependence must be taken into account. Also, because the KRT 3 can only observe the Sun with a high enough flux density for calibration, and the Sun is a variable source not well suitable for calibration, we use a noise generator instead and then perform a ‘Hot–Cold’ measurement.

5.1.1 Determination of the system temperature

The KRT 3 is fitted with a noise generator, which injects a signal directly into the antenna horn. A few preliminary remarks have to be made. The calibration temperature T_{cal} is the expected antenna temperature T_A that is observed in addition when the noise generator is switched on. This temperature can be assumed to be $T_{\text{cal}} = 170 \text{ K}$.⁶ Hence, the system temperature can be estimated from such a Hot–Cold measurement.

During the lab course the Sun is observed, so that we can measure the flux density of the Sun. Then we can check our calibration by comparing this measured values with the current solar flux published everyday online. If one assumes that the Sun is for the KRT 3 is point-like source, then the following applies:

$$T_b \Omega_s = T_A \Omega_A, \quad (3)$$

where T_b is the brightness temperature, Ω_s is the solid angle of the Sun, T_A is the antenna temperature and Ω_A the solid angle of the antenna main lobe.

Preliminary work

We wish to measure the antenna temperature of the Sun and convert this value into a flux density.

- The antenna temperature of the Sun can be easily converted into a flux density. To do this, form equation (3) with the help of the antenna theorem $A_e \Omega_A = \lambda^2$ and the Rayleigh–Jeans law $B_\nu = 2kT_b/\lambda^2$ into an equation that gives the flux density $S_\nu = f(T_A, A_e)$ as a function of antenna temperature T_A and effective antenna area A_e . Furthermore, $S_\nu = B_\nu \Omega_s$ is valid, which must be used;

⁵With the old KRT, for example, $T_h = T_{\text{spillover}} = 20 \text{ K}$ was valid, which was caused by reflections at the antenna components.

⁶We are currently evaluating the noise generator. This value is our best estimate.

- The only unknown size in the equation is the effective antenna area A_e or the aperture efficiency $\eta = A_e/A_g$ with A_g of the geometric antenna surface assumed to be a circle. For the aperture efficiency, we assume $\eta = 0.5$;

Observation

Perform an On–Off measurement with the Sun as the source. At the same time, a Hot–Cold measurement is carried out with the help of the noise generator. To do this, carry out the following steps:

- Point the telescope at the Sun (tab *Source Visibilities*);
- In the *BBC Tools* tab, select all upper side bands (USBs). If you see strong interference, leave out this USB;
- Make the necessary settings for the On–Off measurement (tab *On-Off*):
 1. Repetitions = 5;
 2. Average [s] = 3;
 3. Elev cutoff [°] = 60;
 4. Distance [beam] = 4;
 5. Checkbox *galoff*: Off;
 6. Checkbox *Aquire OnOff Spectra*: Off;
 7. Checkbox *Save as FITS*: Off;
- Then click button Confirm Setup;
- If everything is OK, the button Start OnOff starts the On–Off measurement of the sun.

Then calculate the system temperature T'_{sys} using:

$$T'_{\text{sys}} = T_{\text{cal}} \frac{P_{\text{off}}}{P_{\text{cal}} - P_{\text{off}}}, \quad (4)$$

where P_{off} is the measured power in the OFF position (*off_tpi*) and P_{cal} is the measured power in the OFF position with the noise generator switched on (*cal_tpi*). The units of the measured power are arbitrary.

The antenna temperature of the Sun T_{sun} can be calculated in an equivalent way:

$$T'_{\text{sys}} = T_{\text{sun}} \frac{P_{\text{off}}}{P_{\text{on}} - P_{\text{off}}}, \quad (5)$$

where P_{off} is the measured power in the OFF position (*off_tpi*) and P_{on} is the measured power in the ON position (*on_tpi*). Again, the units of the measured power are arbitrary. The effective system temperature has to be calculated first.

Evaluation

For the evaluation of the data, carry out the following steps:

- Calculate the system temperature in all frequency 16 bands using Equation (4). Average the system temperature separately for the left circular polarisation (BBC 1–8) and the right circular polarisation (BBC 9–16). These values with standard deviation are then the values $T'_{\text{sys,left}}$ und $T'_{\text{sys,right}}$.
- Then calculate the antenna temperature of the Sun with the help of Equation (5) using the calculated system temperature in each BBC. The mean value of the antenna temperature is T_{sun} .
- Research the current flux density S_{ν} of the Sun. The *Space Weather Services* (SWC) of the Australian Government provides current measured flux densities between 245 and 15400 MHz for this purpose.⁷ The published 1415-MHz data can be used in a very good approximation for the KRT 3. Note that the solar flux unit used on the SWC site is defined as $1 \text{ sfu} = 10^{-22} \text{ W m}^{-2} \text{ Hz}^{-1} = 10000 \text{ Jy}$.⁸
- Calculate with the help of the derived formula $S_{\nu} = f(T_{\text{A}}, A_{\text{e}})$ the theoretically expected antenna temperature T_{A} for the Sun and compare this value with the observed value.

5.1.2 Background radiation

Since there is no radio source in the field of view of the antenna at the calibration position you have chosen, the measured antenna temperature is approximately the system temperature $T_{\text{A}} \approx T'_{\text{sys}}$. Measure antenna temperatures T_{A} at different azimuths that are 5 degrees apart. Then measure T_{A} at different elevations (ELV) and fixed azimuth. With RADIOUNIVERSEPRO the easiest way to do this is to perform a cross-scan over a suitable source (other than the Sun). These sources are so weak that they hardly stand out from the background.

Observation

- Find a suitable source (other than the Sun) with an elevation between 30 and 45 degrees (tab *Source Visibilities*);
- point the telescope at this source. To do this, go to the tab *Source Visibilities*. Double-click on the selected *Source*. Wait until the button changes from TRACK to STOP TRACK. Make the settings in the *BBC Tools* tab as before (Section 4.2.2);
- perform a cross-scan (*Total Power Plots* tab) over this source so that the minimum elevation is slightly above 10 degrees. Keep in mind that the span of a total power plot is the entire length of the plot. So half a (*Span*) is observed above and below the source, respectively. A source at an elevation of 40° can thus be observed with a maximum span of 60°, where the elevation of the cross-scan is between 10° and 70°;

⁷https://www.sws.bom.gov.au/World_Data.Centre/1/10#

⁸If the SWC website is not available, alternative data is available from the US National Weather Service's Space Weather Prediction Center (SWPC) via <http://www.swpc.noaa.gov>. The SWPC publishes data at $\lambda = 10.7 \text{ cm}$. You can estimate the flux density of the Sun at $\lambda_{\text{eff}} = 21 \text{ cm}$ using the approximation $S_{\nu} \propto \lambda^{-0.5}$

- generally the span can be calculated like this:

$$Span = 2(ELV - 10^\circ), \quad (6)$$

where ELV is the elevation of the source at the beginning of the observation. If the source sets, i.e. between azimuth values of about 220° – 320° , it is better to take a slightly (2°) smaller span;

- The settings for the tab *Total Power Plots* are as follows:
 1. Reference frame = Horizon;
 2. Span [deg] = as calculated;
 3. Step [deg] = 5;
 4. Average [s] = 3;
 5. Devices: BBCs;
 6. Checkbox *Save as FITS*: OK;
- Then click button *Verify Syntax and System*;
- If everything is OK, the button *START CROSS-SCAN* will start the cross-scan of the source.

Evaluation

Plot the Total Power Intensity (TPI) as a function of offset in degrees. This is done 4 times in total, namely in longitude (azimuth) and latitude (elevation) in both left and right circular polarisation. For this purpose the NASA *fv FITS VIEWER*⁹ the necessary values are exported to a text file. The following steps are taken for evaluation:

- Normalise TPI at the horizontal scan to the calculated system temperature T'_{sys} (Section 5.1.1). Use the same normalisation to plot the vertical scan;
- plot T'_{sys} as a function of azimuth and elevation and discuss qualitatively the variation of T_h . What determines the values of T_h at the different elevations?
- extrapolate the course of T'_{sys} up to an elevation of 0° . Compare the increase in T_h with the Earth's solar radiation temperature of $T_{Earth} = 252 \text{ K}$.¹⁰

5.2 Determination of the angular resolution of the KRT 3

To determine the angular resolution of the KRT 3, the radio telescope is moved step wise over the Sun and the antenna temperatures is measured. If one plots these data against the offset from

⁹<https://heasarc.gsfc.nasa.gov/ftools/fv/>

¹⁰<https://nssdc.gsfc.nasa.gov/planetary/factsheet/earthfact.html>

the Sun, a Gaussian curve is obtained with a half-width that corresponds to the the desired angular resolution. Expected is a half-width $q = 1.02\lambda/D$ rad with the wave length λ and the antenna diameter D . The measurement can be performed with with the help of RADIOUNIVERSEPRO

Observation

Carry out the following steps:

- Align the KRT 3 with the Dun (*Source Visibilities* tab);
- In the *BBC Tools* tab, select all upper side bands (USBs). If a strong interference is seen, omit this USB;
- Note the position calculated on the position of the sun calculated by the control software (e.g. tab *Source Visibilities*);
- Make the necessary settings for the cross-scan in the tab *Total Power Plots* as shown in Fig. 11. The settings are as follows:
 1. Reference frame = Horizon;
 2. Span [deg] = 20;
 3. Step [deg] = 1;
 4. Average [s] = 3;
 5. Devices: BBCs;
 6. Checkbox *Save as FITS*: OK
- Then click button *Verify Syntax and System*;
- If everything is OK, then the button *START CROSS-SCAN* will start the cross-scan over the Sun.

Figure 11: *Cross-scan over the Sun*. Settings for the cross-scan over the sun in the tab *Total Power Plots*.

Evaluation

Record the Total Power Intensity (TPI) as a function of offset in degrees. For this purpose the NASA fv *FITS VIEWER*¹¹ allows one to export the necessary values into a text file.

¹¹<https://heasarc.gsfc.nasa.gov/fv/>

- Fit a Gaussian curve through the measured data points to determine the amplitude;
- normalise the amplitude to the expected antenna temperature of the Sun T_A ;
- the half-maximum width can be expressed by the full width at half-maximum (FWHM) as an angle. Take into account that the angle covered in the case of the longitude measurement does not describe a great circle on the sky but a latitude, so to speak. The angle can be corrected approximately with a factor $\cos(ELV)$, where ELV is the elevation in degrees;
- the resolution is determined 4 times, namely in longitude (azimuth offset) and latitude (elevation offset) for the left- and right-circular polarisations;
- compare the obtained resolution with the theoretical value and theoretical value and discuss possible deviations.

5.3 Determination of the sensitivity of the instrument

5.3.1 Theoretical sensitivity

The weakest detectable flux density ΔS_V for a given system temperature T'_{sys} can be given by the radiometer equation:

$$\Delta S_V = \frac{2k}{A_e} \frac{C_s T'_{sys}}{\sqrt{\Delta\nu \tau}}, \quad (7)$$

where ΔS_V is the (spectral) flux density in $\text{W m}^{-2} \text{Hz}^{-1}$, $k = 1.38 \times 10^{-23} \text{ J K}^{-1}$ is the Boltzmann constant, A_e is the effective antenna surface, $C_s \approx 2$ a dimensionless instrumental constant, $\Delta\nu$ is the bandwidth, τ is the integration time and T'_{sys} the system temperature.¹²

Evaluation

Carry out the following steps to determine the theoretical limiting sensitivity ΔS_V of the instrument:

- Calculate the limiting sensitivity ΔS_V in a single channel ($\Delta\nu = 61 \text{ kHz}$) for the case where 1 second is integrated;
- find a bright radio source (either Cyg A, Cas A or Tau A) suitable for observation, which has an elevation of ideally $ELV \geq 40^\circ$. You can check this in the *Source Visibilities* tab;
- calculate the necessary integration time τ to detected the selected source with the KRT 3 with a signal-to-noise ratio $S/N = 10$ in each channel. Use the flux densities in Table 2.

¹²Modern radio telescopes are perfected in all their components to the point to the point where $C_s \approx 1$. The value for the KRT 3 is an estimate.

Table 2: Flux densities of some bright radio sources at $\nu = 1420.4$ MHz. An uncertainty of 5 per cent was assumed. After Baars et al. (1977).

Source	Sourcename	Flux density (Jy)	Nature of the Source
Cas A ¹	Cassiopeia A	1420 ± 70	Supernova remenant
Cyg A	Cygnus A	1560 ± 80	Radiogalaxy
Tau A	Taurus A	940 ± 50	Supernova remenant
Vir A	Virgo A	211 ± 11	Radiogalaxy

¹ At epoch 2021.0; the flux density of Cas A decreases by 0.924 per cent each year.

5.3.2 Determination of the uncertainty of radiant flux measurements

Flux density measurements are always performed relatively, i.e. a measurement with the source (ON) is compared with a measurement without the source (OFF). The change between ON and OFF position position takes place in such short intervals that the observation observation conditions can be regarded as constant. The weather conditions may vary as may be the quality of the tracking and the sensitivity of the receiver. Typical integration times per position are in the range of minutes or shorter.

To separate the background from the antenna temperature, a ‘Beamswitch’ measurement is performed, in which the position of the source (ON-position) and a reference position next to the source (OFF position) is measured alternately. If an On–Off measurement is recorded (tab *On-Off*), RADIOUNIVERSEPRO stores the antenna temperatures $T_A(ON)$ and $T_A(OFF)$. From the difference one obtains the antenna temperature of the source T_s :

$$T_A(ON) - T_A(OFF) = (T_s + T'_{sys}) - T'_{sys} \quad (8)$$

Preliminary work

- The radio signals from space received on earth are weak. For comparison, calculate how great the flux density of a mobile radio of a mobile phone at 900 MHz on Earth, transmitting isotropically with power of 2 W and is located on the moon. Assume that the spectrum of the mobile radio transmitter has a normal distribution around 900 MHz with a half-width FWHM¹³ = 80 kHz. Compare the value with the flux density of Cas A at 900 MHz. Calculate the flux density of Cas A under the assuming that the spectral response is a simple power law of the form $S_\nu \propto \nu^\alpha$ with $\alpha = -0.7$;
- Using the equation $S_\nu = f(T_A, A_e)$ you can calculate the antenna temperature T_A expected for a given source. Calculate this temperature for the source Cas A.

Observation

¹³Note that for the application of a normal distribution, the half-width FWHM must be converted into the standard deviation σ

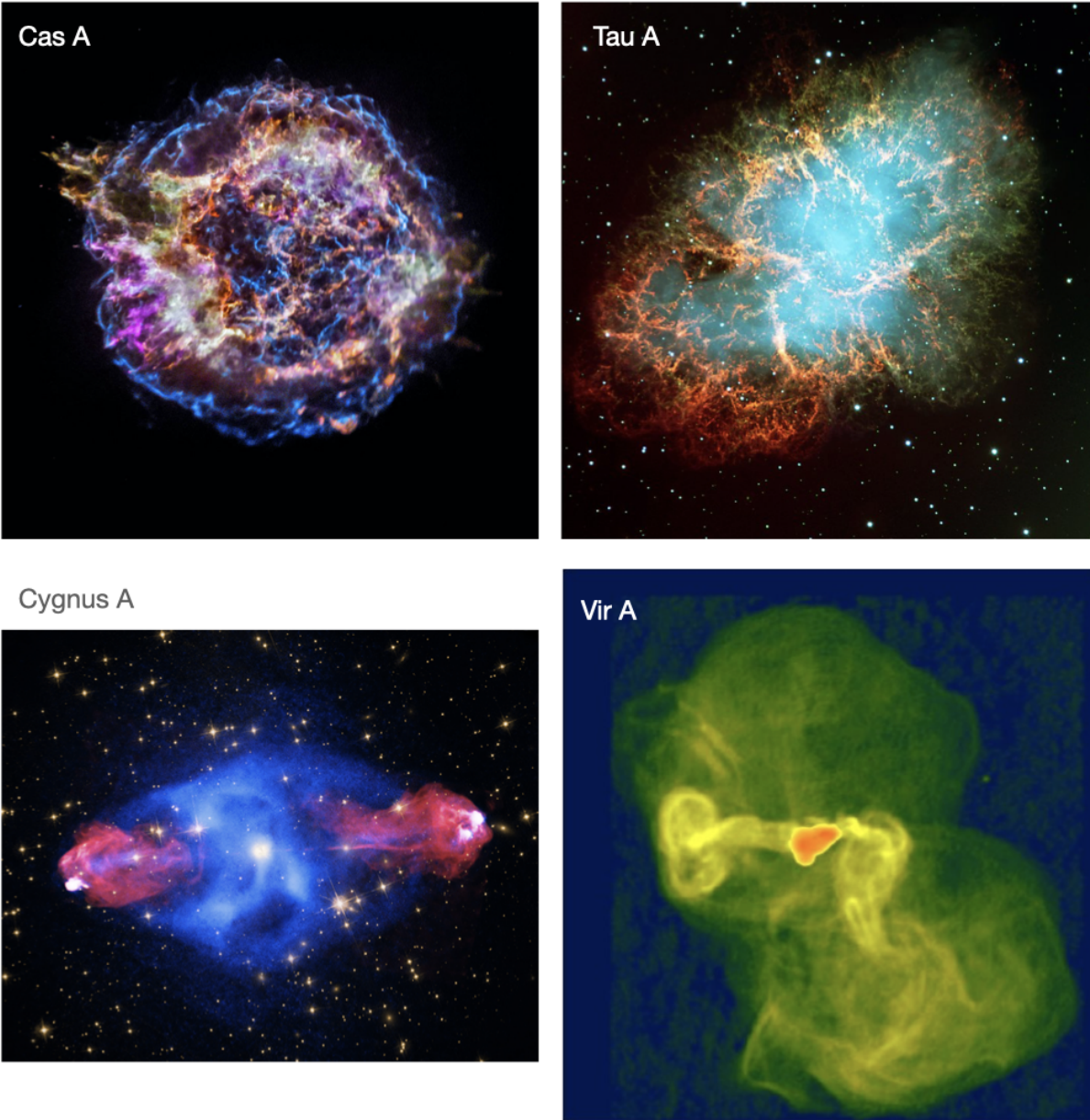


Figure 12: *Some bright radio sources.* Xray image of Cassiopeia A (source: NASA), optical image of Taurus A (source: ESO), radio and Xray image of Cygnus A (source: NASA and NRAO) and radio image of Virgo A (source: NRAO).

- Determine the antenna temperature of a source (Cyg A, Cas A or Tau A) by mapping the source. Instead of Cyg A, the massive starburst region W75N can also be observed. The observation consists of a grid pattern on the sky where the KRT 3 takes measurements line by line at different positions. Different columns are also scanned, which allows the grid pattern to be assembled into a map. The number of map pixels is $\text{pix} = \text{rows} \times \text{columns}$, where *rows* is the number of rows and *columns* is the number of columns;
- In the *Source Visibilities* tab, check which sources are good to observe at the moment.

For mapping, sources with elevations between 30° and 60° are ideal. Point the telescope at the source. Wait until the button changes from TRACK to STOP TRACK;

- In the *BBC Tools* tab, adjust the gain again and select only the frequency bands that do not show strong RFI influence. Also omit frequency bands with the H I line;
- Switch to the tab *Mapping*. Make the following settings (see Fig. 13):
 1. Width = 10 deg; Height = 10 deg;
 2. Delta = 0.5 HPBW;
 3. Average = 10 s;
 4. Confirm the button Confirm Setup;
 5. If everything is OK, pressing the button Start Map will start the mapping.



Figure 13: *Mapping of a Source*. Settings for mapping a source in the tab *Mapping*.

- Note the time needed for mapping;
- Choose a suitable colour coding for the map using the *False colours palette* option. Also test *log* and *square root* scaling, as well as the *Pixelise* or *smooth* option in the *Processing* section;
- When you are happy with the way the map looks, it will load itself with the button Save as bitmap. However, it is recommended to also take a *screen-shot* so that the axis labels are preserved. An example of a map can be found in Fig. 14.

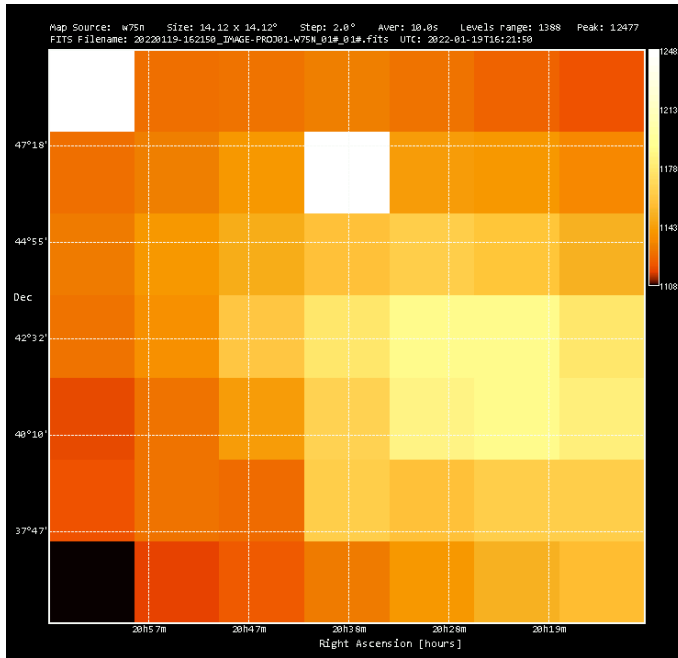


Figure 14: *Map of the star-forming region W75N.* The map shows the radio continuum emission at 1420 MHz with a resolution of 4.5° FWHM. The map size is about 14 × 14 square degrees. The intensity is in uncalibrated units.

Evaluation

- Measure the mean and standard deviation of the map in the uncalibrated units. This can be done with the software *ds9*,¹⁴ use a sufficiently large (at least 20 pixels) area of the map in which you position a box (Menu Region → Shape → Box);
- Now normalise the mean to the system temperature T'_{sys} , which is then also our $T_A(\text{OFF})$. Then you can also determine the map noise in units of Kelvin with this scaling. Now one still determines the antenna temperature at the source position $T_A(\text{ON})$ by converting the measured value in *ds9* again with the scaling;
- Calculate the antenna temperature of the source using equation (8);
- What can you tell about the background? Is constant? Discuss the results quantitatively by determining the value of S/N ;
- From the log you can get the actual measurement times. What percentage of the measurement time is accounted for by the observation of the source (ON measurement), if 10 seconds are integrated per measurement point? How many maps do you use to detect the source with a signal-to-noise ratio $S/N = 10$ for the source? Assume the ideal case where $S/N \propto \sqrt{\tau}$ holds. Compare this result with the theoretical expected value. What conclusions do you draw from the mapping of the source and what influences the quality of the measurement significantly?

¹⁴<https://sites.google.com/cfa.harvard.edu/saoimageds9>

5.4 Observation of neutral hydrogen in the Milky Way

5.4.1 Determine the frequency setting of the receiver

The preamplifier has a frequency range 1398.00–1460.56 MHz divided into 1024 channels. The frequency bands of the receiver can be set to any frequency in the frequency range of the preamplifier. Preset bandwidths are 8 MHz each, where the frequency set in *BBC Tools* refers to the lower limit of the range. The laboratory frequency of the 21cm line of HI is 1420.4 MHz. The hydrogen clouds in the Milky Way usually have a radial velocity of $v_{\text{rad}} \neq 0$ above the observer on Earth, i.e. the frequency of the emitted by them is shifted with respect to the laboratory frequency.

Preliminary work

- What is the frequency resolution $\delta\nu$, i.e. the bandwidth of a single channel? Then what is the velocity resolution δv when observing the HI line?
- In which channel would you expect the HI line at $\nu_0 = 1420.4$ MHz, assuming a radial velocity of 0 km s^{-1} ?
- You want to measure the radio continuum of a source. What frequencies are allowed within the bandwidth so that the measurement is not be contaminated by the 21cm line. Assume a maximum radial velocity of the HI of $\pm 200 \text{ km s}^{-1}$;
- So which frequency bands should be selected in *BBC Tools* if you want to observe the HI line once and the continuum once?

5.4.2 Measurement of the 21cm line along the Milky Way plane

Hydrogen is the most abundant element in the cosmos and accounts for about 80% of the mass of visible baryonic matter. With the help of the 21cm line, hydrogen can be detected throughout the galaxy. It is formed by a transition in the ground state of the neutral atom (HI). While interstellar dust blocks the view of large parts of the Milky Way for optical wave propagation, it is optically transparent for radio waves. The Doppler effect allows the radial velocities of the gas along the line of sight to be determined. Typical temperatures in interstellar HI clouds are $\approx 100 \text{ K}$, so that the thermal broadening of the lines is $\approx 1 \text{ km s}^{-1}$. The observed HI line profiles show a velocity range that is in part more than 100 km s^{-1} large, which is therefore due to different rotational velocities of the HI-gas within the Galaxy.

In the first approximation, it is assumed that the hydrogen in the plane on circular orbits around the Galactic centre. Then the orbital velocity $v(R)$ is valid:

$$v(R)^2 = \frac{GM_R}{R}, \quad (9)$$

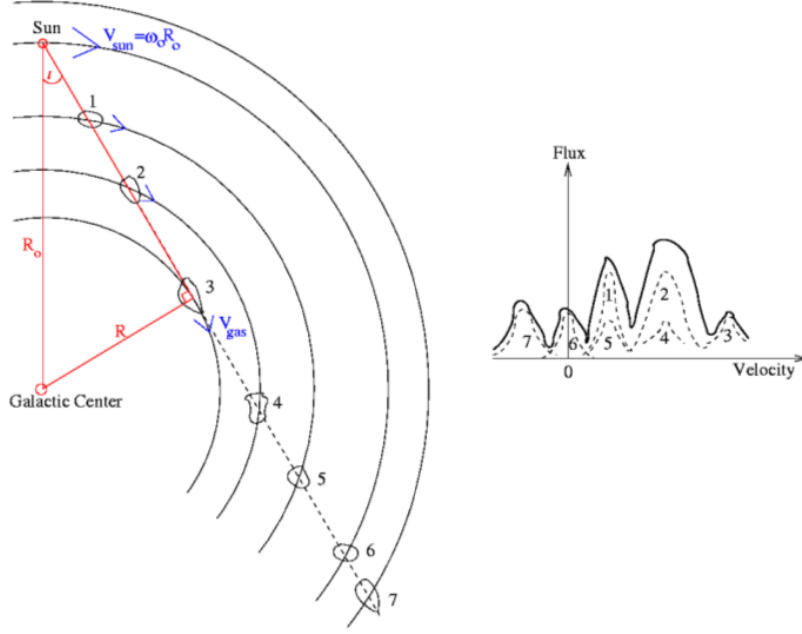


Figure 15: Schematic representation of the rotation of HI clouds around the Galactic centre and the resulting line profile. The profile is formed by superposition of the emissions of individual clouds with different radial velocities.

with M_R as the mass of the Milky Way within the radius R and the gravitational constant $G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$. The different HI-clouds therefore have different velocities depending on their distance from the Galactic centre.

If one observes along a line of sight with the angle l (Galactic length) with respect to the line connecting the Sun and the Galactic centre (Fig. 15), the observed HI profile is an overlay of the emissions of all the HI clouds along these lines of sight. Due to their different radial velocities above the Sun, the HI-profile extends over a large velocity range. If the orbital velocities do not increase at greater distances from the Galactic centre, then the highest radial velocity v_{rad} is reached at the *tangential point*, i.e., at the point where the line joining the Galactic Centre forms a right angle with the visual beam. The distance R of the tangential point from the Galactic centre is $R = R_0 \sin l$, with R_0 being the distance of the Sun from the Galactic Centre ($R_0 = 8.5 \text{ kpc}$, $1 \text{ kpc} = 3.086 \times 10^{19} \text{ m}$).

Let the orbital velocity at the tangential point be $v = \omega R$, with ω the angular velocity at distance R . The orbital velocity of the sun is $v_\odot = \omega_0 R_0 = 220 \text{ km s}^{-1}$, with ω_0 the angular velocity at distance R_0 . For the radial velocity v_{Rad} between the tangential point and the sun, the following holds true

$$\begin{aligned} v_{\text{rad}} &= \omega(R) R - \omega_0 R_0 \sin l \\ &= v(R) - \omega_0 R \end{aligned} \tag{10}$$

Table 3: Transit time of the Galactic Centre for the Hamburg Observatory during the year for the 15th day of each month.

Monat	Transit
Januar	10:30 ¹
Februar	08:30 ¹
März	06:30 ¹
April	05:30 ²
Mai	03:30 ²
Juni	01:30 ²
Juli	23:30 ²
August	21:30 ²
September	19:30 ²
Oktober	17:30 ²
November	14:30 ¹
Dezember	12:30 ¹

¹ MEZ; ² MESZ

However, the radial velocity v_{obs} , which is calculated from the frequency shift of the 21cm line measured at KRT 3, still contains velocity components due to the rotation of the Earth and its orbit around the Sun. These must be corrected to obtain v_{rad} . In addition, corrections are made for the motion of the solar system itself to arrive at a reference system for velocities that does not move relative to the rotation of the Milky Way disk. This system is called 'local-standard-of-rest' (LSR). The correction velocity Δv_{LSR} is calculated by the PYTHON script `spectrum.py` and output in the terminal. The radial velocity v_{LSR} with respect to the LSR is obtained from $v_{\text{LSR}} = v_{\text{obs}} - \Delta v_{\text{LSR}}$. The radial velocities that '`spectrum.py`' calculates are already corrected and in relation to the LSR. So then one only needs to determine the maximum value of v_{LSR} , which then simultaneously represents the radial velocity v_{rad} at the tangential point.

Take spectra at Galactic length $l = 10^\circ, 20^\circ, 30^\circ, \dots, 90^\circ$. Because the Galactic centre is at a declination of $\delta_{\text{GZ}}(\text{J2000}) = -29^\circ$, it only reaches a maximum height of 7° above the horizon as seen from Hamburg. Accordingly, you have to adjust the observation time so that you observe it at the maximum altitude. This is the case when the source is exactly in the south, i.e. just in the *transit*. Since the transit time shifts during the year and therefore does not coincide with the working time, alternative galactic lengths may have to be observed. The Galactic Centre is located at a right ascension of $\alpha_{\text{GZ}}(\text{J2000}) \approx 17^{\text{h}}45^{\text{m}}$. Therefore, it is optimal to observe at a local stellar time (LST) of $\text{LST} = 17:45:00$. This means that the ideal observing time extends over the whole daytime in the course of the year (see Table 3).

Observation

Perform an On–Off measurement at a point on the Galactic plane. Sources are controlled with Galactic coordinates (l, b) , where l is the Galactic length and b is the Galactic latitude. The Galactic centre is at $l = 0^\circ$, $b = 0^\circ$; the choice of $b = 0^\circ$ means that the point is in the Milky Way plane.

- Align the telescope to the one point in the Galactic Plane. To do this, use the com-

mand source=gal,90,0 in the command line of RADIOUNIVERSEPRO (grey line Left below). This command will move the telescope to a Galactic Length of $l = 90^\circ$ and a Galactic Latitude of $b = 0^\circ$. Wait until the button changes from TRACK to STOP TRACK;

- In the *BBC Tools* tab, select the appropriate upper side bands (USBs) to observe the H I line;
- Make the necessary settings for the On–Off measurement. The settings are as follows:
 1. Repetitions = 10;
 2. Average [s] = 10;
 3. Elev cutoff [$^\circ$] = 60;
 4. Distance [beam] = 4;
 5. Checkbox *galoff*: OK;
 6. Checkbox *Aquire OnOff Spectra*: OK;
 7. Checkbox *Save as FITS*: OK.
- Then click on button Confirm Setup;
- If everything is OK, the button Start OnOff starts the On–Off measurement of the Milky Way.

During the observation, examine the spectra as they build up during the on-off measurement (tab *Spectrometer*, see Fig. 16). Zoom in on the H I line and take screenshots of the spectra (left click and hold allows to drag a window open, the small icon at the bottom left of each spectrum allows to zoom back in). With small Galactic length, part of the H I line shifts to lower and lower frequencies. Therefore, the zoom window may need to be adjusted. Move the window so that the centre of the line is centred. The lowest achievable galactic elevation depends on the time of observation. The lowest allowed elevation of the telescope is elevation = 10° .

Evaluation

- Averaging the spectra that belong to a given Galactic length. To do this, use the PYTHON script `spectrum.py`. Plot one of the spectra with $l \geq 30^\circ$ as a function of frequency. Compare this with the control spectrum generated by the control software during the observation. Discuss any discrepancies;
- Compare this spectrum as a function of radial velocity with the H I-line profile of the Leiden-Argentine-Bonn (LAB) survey¹⁵ at this sky position. Download the spectrum as an ASCII file and plot both spectra on top of each other.¹⁶

¹⁵https://www.astro.uni-bonn.de/hisurvey/AllSky_profiles

¹⁶Note that one must specify the correct resolution for the LAB survey. However, the observed half-width of the lobe from section 5.1 is larger than the allowed value of ‘Effective beamsize FWHM [$^\circ$]’ (in units of degrees), which should be ≤ 1 . So you just have to enter 1.0 as the value. Since the antenna temperatures are surface brightnesses, this is a useful approximation for extended sources.

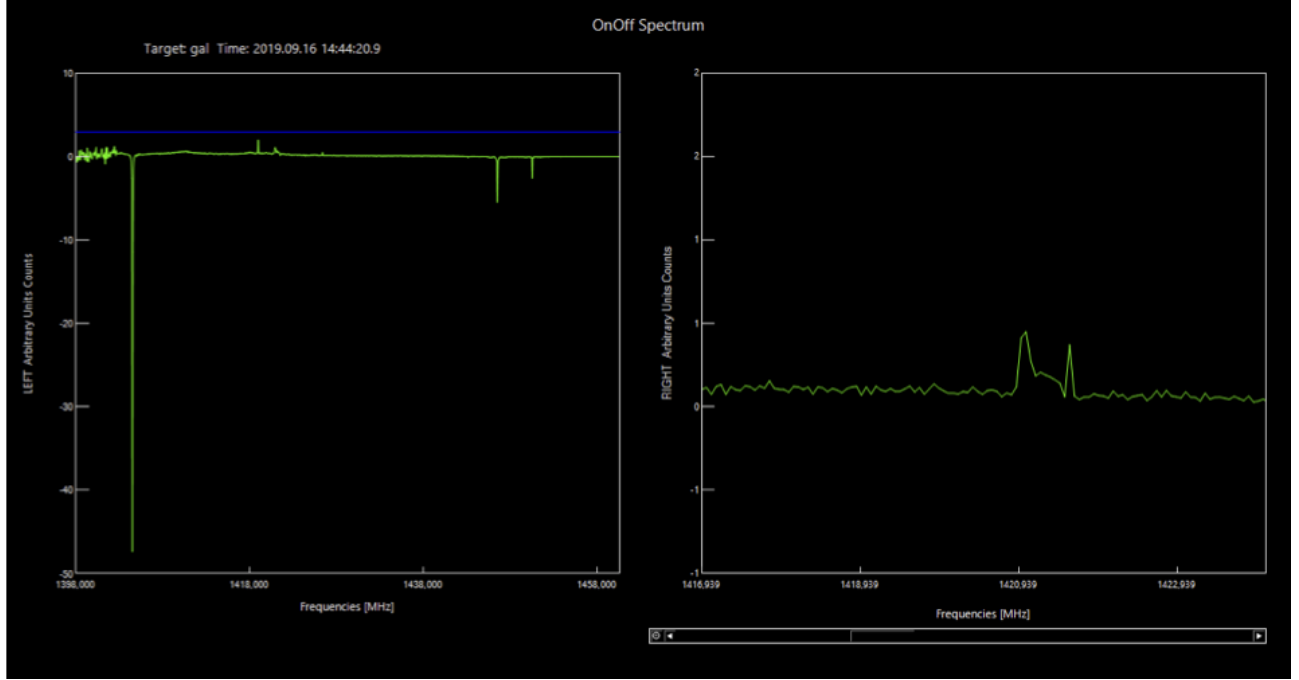


Figure 16: Control spectrum of the H I line in the *Spectrometer* tab. This spectrum is continuously updated during the on-off measurement. It can be viewed interactively.

Here, the KRT 3 measurements should be shown as data points with error bars and the LAB measurements as a solid line. The LAB measurements here represent an idealised model, so to speak. Discuss deviations;

- In all averaged spectra, define the channel with the maximum velocity of the H I-profile. To do this, place a divided baseline through the spectrum; this is a line with a constant temperature constant temperature T_A , which represents the region of the spectrum with no emission. no emissivity as closely as possible. In this step obtain the maximum radial velocity v_{rad} together with an estimate of the inaccuracy;
- Calculate the orbital velocities $v(R)$ at the tangential points $v(R) = v_{\text{rad}} + \omega_0 R$;
- Generate a rotation curve in which you graphically plot the the orbital velocities of the tangential points against their distance from the galactic centre. What do you expect the curve to look like?
- Calculate the enclosed mass of the Milky Way (in solar masses) at each distance R and also plot this graphically. You can assume, in useful units:

$$M_R = 0.234 \times 10^{10} \left(\frac{v(R)}{100 \text{ km s}^{-1}} \right)^2 \left(\frac{R}{\text{kpc}} \right) M_{\odot}, \quad (11)$$

where $1 M_{\odot} = 1.98 \times 10^{30} \text{ kg}$ is one solar mass.

Discussion

Here are some hints for a possible discussion of the results from the measurements of the H I spectra.

- How does the measured rotation curve compare with literature data? An example is given in the introduction. Discuss possible deviations.
- When comparing the H I spectra with the structure of the Milky Way (Figure 1), are there any discrepancies?
- The total stellar mass of our Milky Way is about 100 billion solar masses. How does that compare to the enclosed mass within the solar orbit? What can we conclude about the entire Milky Way?
- The total mass of the Milky Way within a radius of 100 kpc is about 10^{12} solar masses. How does that compare with your results within the solar orbit?

5.4.3 Distribution of dark matter

The rotation curve of the Milky Way shows the total mass distribution consisting of ordinary visible (baryonic) matter and dark matter. The visible matter is concentrated in a relatively thin disk, while the dark matter surrounds the Milky Way in a spherical halo. To calculate the distribution of dark matter, the mass distribution of baryonic matter (stars and gas) can be subtracted. The surface mass density of the stellar and gaseous disk can be described approximately as an exponential function $\Sigma_d = 1905 \exp(-R/2 \text{ kpc}) \text{ M}_\odot \text{ pc}^{-2}$ (Binney and Tremaine 2008). Thus, the corresponding orbital velocity can be calculated as follows:

$$v_d = 452 \sqrt{\frac{\text{kpc}}{R} \left[1 - \left(\frac{R}{2 \text{ kpc}} + 1 \right) \exp\left(\frac{-R}{2 \text{ kpc}}\right) \right]} \text{ km s}^{-1} \quad (12)$$

Extend your plots from section 5.4.2 to calculate the distribution of dark matter:

- Enter the shape of the baryonic rotation curve from equation (12) into your measured rotation curve;
- With this, you can now also show the fraction of baryonic mass $M_d(R)$ in the mass distribution. After subtracting the baryonic matter, one obtains the mass distribution $M_{\text{DM}}(R)$ of the dark matter. Determine the mass $M_{\text{DM}}(R_0)$ of dark matter enclosed within the solar orbit;
- Assuming a spherical distribution for the dark matter, i.e. $M_{\text{DM}}(R_0) = 4/3\pi\langle\rho_{\text{DM}}\rangle R_0^3$, what is the mean density $\langle\rho_{\text{DM}}\rangle$ of the dark matter within the solar orbit? Compare this value with the local dark matter energy density to $\rho_{\text{DM}}(R_0)c^2 = 0.43 \pm 0.11 \text{ GeV cm}^{-3}$ (Salucci et al. 2010).¹⁷

¹⁷ $1 \text{ GeV cm}^{-3} = 1.6 \times 10^{-4} \text{ J m}^{-3}$

6 Conclusion of the observations

Once you have finished your observations with the KRT 3 please do the following the following steps:

- Point the telescope at the sun if possible (section 4.2). Check the pointing correction. This measurement is to verify that the pointing correction has remained stable during the measurements;
- Move the telescope to the parking position: button Park select. If this is not possible (e.g. only the button Unpark is visible), the command `source=azel,270,90` will return the KRT 3 to the park position (Azimuth = 270°, Elevation = 90°);
- After reaching the rest position, first switch off the receiver. To do this, press the button Stop in *Receivers and Data Acquisition Control* (top right). Then press the button Disconnect button;
- Next, switch off the mount. To do this, press the button Disconnect in the *Mount Control* area;
- Then exit the control screen in the *File* menu by pressing the *Exit* option;
- Exit the VLC media player;
- Copy all the files you have created to a folder on the observatory's Nextcloud. The files on the lab computer will be deleted without warning after a few weeks. You are responsible for the integrity and storage of your data;
- Log out of the computer.

7 Evaluation

7.1 Data reduction

The easiest way to do the data reduction is to use PYTHON. As a result of the measurements you have several FITS files *filename.fits* containing the recorded On-Off HI spectra. The files can be read and processed with the PYTHON script `spectrum.py`. The FITS header contains all relevant data such as the date given as Julian date (JD). It can be converted to a common date, for example, here¹⁸. The FITS-Header can be read with the NASA FITS viewer `fv` (Fig. 17).

For a quick analysis of the data, the spectra can be displayed graphically with `spectrum.py`. Figure 18 shows an example.

¹⁸<https://www.aavso.org/jd-calculator>

fv: Binary Table of l20.fits[1] in /home/hslxrsrv3/stnv047/det/

File Edit Tools Help

Select	JD 1D d	Azimuth 1D deg	Elevation 1D deg	Az_Offset 1D deg	El_Offset 1D deg	Gal_Long 1D deg
All	Modify	Modify	Modify	Modify	Modify	Modify
1	2.458718394713E+06	2.078100000000E+02	2.126000000000E+01	-1.559789844822E-02	3.468471756509E-03	2.035615087905E+01
2	2.458718395256E+06	2.252900000000E+02	2.346000000000E+01	1.727226707446E+01	2.259344952477E+00	2.034534099148E+01
3	2.458718395469E+06	2.253700000000E+02	2.343000000000E+01	1.727651674870E+01	2.260394472004E+00	2.035926692795E+01
4	2.458718396054E+06	2.082900000000E+02	2.113000000000E+01	-1.375643428645E-02	9.139018784975E-03	2.036132063153E+01
5	2.458718396661E+06	2.257900000000E+02	2.324000000000E+01	1.726778534405E+01	2.190659238774E+00	2.035002779068E+01
6	2.458718396869E+06	2.258700000000E+02	2.322000000000E+01	1.726221900205E+01	2.182055630603E+00	2.035248280430E+01
7	2.458718397463E+06	2.088000000000E+02	2.098000000000E+01	-1.173493005371E-02	2.959942596796E-03	2.035484493490E+01
8	2.458718398010E+06	2.262700000000E+02	2.303000000000E+01	1.725291188707E+01	2.111866475196E+00	2.033896263056E+01
9	2.458718398213E+06	2.263400000000E+02	2.301000000000E+01	1.724752794188E+01	2.113601521283E+00	2.035200742901E+01
10	2.458718398753E+06	2.092700000000E+02	2.085000000000E+01	-1.499287251892E-02	9.380232690958E-03	2.035975681253E+01
11	2.458718399286E+06	2.267200000000E+02	2.283000000000E+01	1.723266936287E+01	2.045493978688E+00	2.033788583595E+01
12	2.458718399474E+06	2.267900000000E+02	2.281000000000E+01	1.724163444398E+01	2.046316265200E+00	2.034833372246E+01
13	2.458718400000E+06	2.097200000000E+02	2.071000000000E+01	-1.627795180630E-02	1.165397142518E-02	2.036215917447E+01
14	2.458718400529E+06	2.271500000000E+02	2.264000000000E+01	1.722181607253E+01	1.988550577112E+00	2.034423585795E+01
15	2.458718400711E+06	2.272200000000E+02	2.262000000000E+01	1.721927795646E+01	1.987175611564E+00	2.035191355033E+01
16	2.458718401241E+06	2.101600000000E+02	2.058000000000E+01	-2.238293101993E-02	4.512057202213E-03	2.035607731235E+01
17	2.458718401756E+06	2.275800000000E+02	2.244000000000E+01	1.721030209404E+01	1.920921928770E+00	2.033791275875E+01
18	2.458718401926E+06	2.276500000000E+02	2.242000000000E+01	1.721206257426E+01	1.928536044369E+00	2.035398381960E+01
19	2.458718402468E+06	2.106100000000E+02	2.045000000000E+01	-2.332975330268E-02	7.892095921999E-03	2.035947856731E+01
20	2.458718402979E+06	2.280100000000E+02	2.225000000000E+01	1.719392614538E+01	1.863770133438E+00	2.034168277739E+01

Go to: Edit cell:

Figure 17: Display the FITS-Headers of one of the output files with the NASA Fits Viewer fv. For example, you can find out the date of the observation (Julian date; JD), azimuth, elevation, and Galactic latitude.

To do this, the following command is used in the shell:

```
$ python spectrum.py l110.fits
```

The default is then to plot the antenna temperature on the y axis and the radial velocity with respect to the LSR on the x axis. The first step is to subtract a baseline, assumed to be 10 K. The command for this is:

```
$ python spectrum.py l110.fits vlrs 10
```

The option vlrs tells the programme that the x-axis should be the radial velocity with respect to the LSR. The option '10' is then the constant baseline of 10 K that is subtracted. In order to be able to plot the LAB data for comparison for the calibration of the H I data, this can be specified as an additional option:

```
$ python spectrum.py l110.fits vlrs 10 lab_l110.txt
```

Here the file lab_l110.txt is read in and plotted as a solid line.

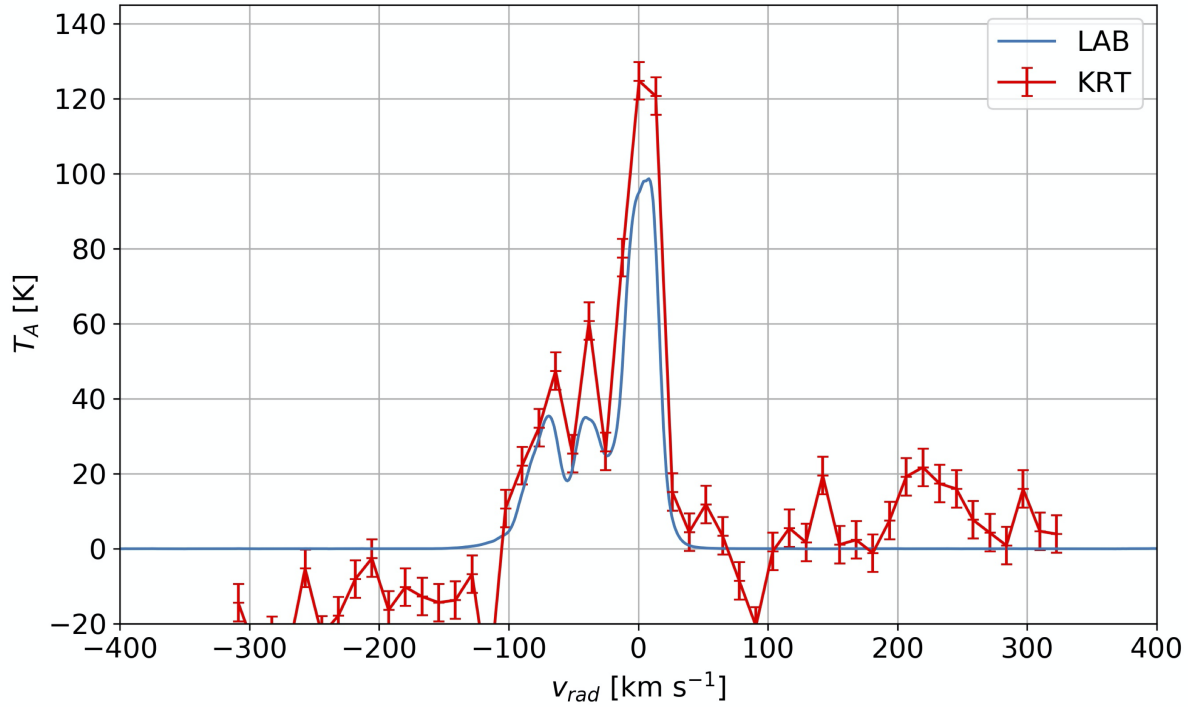


Figure 18: Section of a H I spectrum after processing the FITS file with `spectrum.py`. The x-axis shows the radial velocity in km s^{-1} with respect to the LSR and the y-axis shows the antenna temperature in Kelvin. Data points with error bars are measured from KRT3 and the solid blue line shows a cross-measurement of the LAB survey.

7.2 Radio frequency interference (RFI)

Although the area around the H I line is a protected frequency range, some terrestrial interference occurs which may affect the sensitive measurements. Corrections must therefore be made for any erroneous frequency channels that are strongly affected by radio frequency interference (RFI). (RFI). This can be recognised in the spectra by the fact that an artificial line (RFI) has a width of only one channel. The faulty values can be replaced e.g. by the mean value of the adjacent channels. Alternatively, it may be that the entire spectrum is affected by RFI, but only a subset of the On–Off measurements, i.e. RFI limited in time, occurs.

The identification of erroneous data can be done by the following methods

- Plot of antenna temperature vs. frequency channel of individual (or all) spectra. Do the spectra look reasonable? For radio continuum measurements of the Sun, the antenna temperature should be nearly constant in the sensitive area of the filter. This can be done with the following command:

```
$ python spectrum.py l110.fits chan 10
```

Now plot the antenna temperatures as a function of channels, again with a baseline of

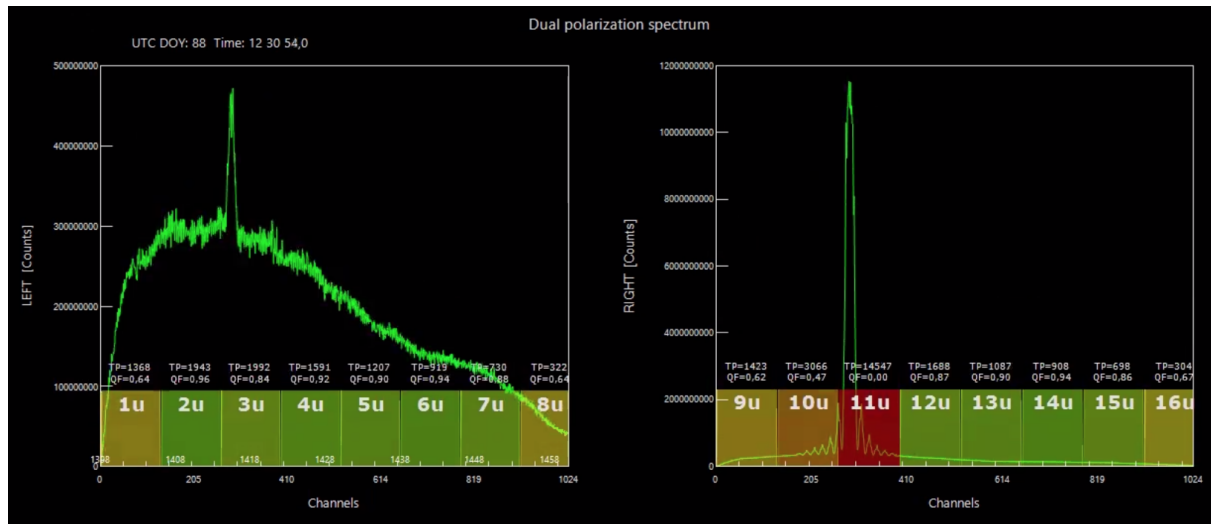


Figure 19: Example of a strong radio frequency interference (RFI) signal. In most cases the RFI signals are very narrow in their spectral (frequency) distribution and are also highly variable in time, so that they are easy to recognise as interference signals. In the example shown, the RFI signal is also strongly polarised, because it is much more prominent in the right polarisation. This is typical for RFI signals, whereas astronomical sources are usually only weakly polarised.

10 K subtracted;

- Plot of antenna temperature vs. frequency. Similar to before, you can also plot antenna temperature as a function of frequency:

```
$ python spectrum.py l110.fits freq 10
```

- Averaging the antenna temperatures over the sensitive range of the filter and plotting the mean against time. This can be done by looking at the spectra of the 10 On–Off measurements individually. If a measurement is strongly influenced by RFI and has a high rms noise, one should omit this measurement. To do this, the line:

```
observations = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
```

in the *spectrum.py* script so that only the useful observations remain. Remember that PYTHON starts counting from 0, so if the On–Off measurement 4 is strongly influenced by RFI, then one must omit the ‘3’ in the observations.

Decide if the data is usable or if it is necessary to repeat the observation.

7.3 Graphic representation and analysis

Use a graphics programme of your choice for graphical representations. See the appendix for examples of graphing with PYTHON.

For pure continuum measurements you can use the antenna temperatures averaged over all channels. You can use the programme `fv` to convert the FITS files into text files. This is mainly done for the cross-scans over the background source and over the sun. As a result you get mean antenna temperatures in the left and right circular polarisation. These can be plotted against azimuth and elevation.

For On–Off (Beamswitch) measurements, the mean antenna temperatures of the OFF measurements must be subtracted from the ON measurements. This is the only situation where reading the data from the screen is easiest. These data can then be evaluated with a tabulation.

For spectra (H I-line) the averaging has to be done over each frequency channel to get an average spectrum that can be plotted. Think about a criterion for finding the frequency channel with the maximum H I.

8 Experiment protocol

Write the experimental protocol in the form of an astrophysical publication. It should contain the headings *Introduction*, *Observation and Evaluation*, *Results*, *Discussion* and *Appendix*. The recommended length is about 15-20 pages (plus appendix). You can write the experimental protocol with a word processor of your choice. The use of $\text{\LaTeX}$ is recommended; the finished protocol should be submitted in PDF format. A possible structure is as follows:

Introduction (1–2 pages): Describe the aim of the practical experiment and the procedure to achieve the aim. The objective should be the rotation curve of the Milky Way in order to derive the mass distribution. Radio observations are particularly suitable for this purpose because, unlike optical observations, absorption by dust plays only a minor role. In addition, radio observations of the Earth's surface are possible, which is otherwise only the case with optical observations. Here we can also briefly discuss the emission process behind the 21cm radiation;

Observation and Evaluation (3–5 pages): Describe the instrument used, the general settings used, the data recorded and your evaluation method. The date of the observations and the weather conditions belong here as well as the results of the pointing corrections. The preparatory questions should also be answered here;

Results (5–10 pages): For each of the experimental parts from section ??, describe the results you found. In particular, all plots and graphs should be described. If data were fitted, the results should be described here;

Discussion (2–4 pages): In this section, you will conduct an evaluation of your results of each part of the experiment. Were the objectives met, are the results in line with expectations? If necessary, discuss the influence of external factors, such as the influence of background radiation, pointing accuracy, and the choice of baseline for the H I measurements;

Appendix: Attach a copy of your measurement report, as well as additional material if desired. The measurement report can be in an unedited raw form, a photo of the handwritten transcript is sufficient. All H I spectra should also be shown in the Appendix.

A few general notes on the preparation of the protocol:

- errors are usually only given to a maximum of 2 digits. The result has to be adjusted accordingly. So e.g. $T'_{\text{sys}} = 212 \pm 12$ K instead of $T'_{\text{sys}} = 212.02 \pm 12.45$ K;
- The axis labels in the plots shall be adjusted so that the font size is at least as large as that of the caption;
- Units are not slanted with a space after the result. In composite units, distances must also be observed, e.g. $v_{\text{rad}} = 100 \text{ km s}^{-1}$ instead of $v_{\text{rad}} = 100 \text{ km s}^{-1}$;
- For decimal places the dot is recommended instead of the comma, e.g. $T_s = 2.1 \pm 0.4$ K instead of $T_s = 2,1 \pm 0,4$ K, so that there are no differences between the numbers;
- The axis labels in figures should have a font size that is not much smaller than the figure captions or any other small font size in the text. Keep in mind that they have to be readable when printing out the document (although this rarely done nowadays but this is convention).

A Troubleshooting in KRT 3 operation

A.1 General notes

The KRT 3 is mechanically very stable. If faults should occur, this is usually due to RADIOUNIVERSEPRO which is still under development. It is important to note the *elevation limit* of $ELV = 10^\circ$. The KRT 3 produces an error if you want to observe below it. It is also recommended to always use the *arrow keys* or *drop-down menus* to enter or change numbers in the RADIOUNIVERSEPRO. Unfortunately, entering numbers manually via the keyboard sometimes causes the software to crash. It often happens that the KRT 3 does not follow up on a source. Then the error message ‘not tracking’ appears in the log window. In this case it is usually enough to align the telescope to the source again. Double-click on the source in the *Source Visibilities* tab. The control software can also lose the position information of the telescope if you observe over a longer period of time. Then you have to repeat the alignment with the Sun.

The software REMMINA sometimes loses the connection to the desktop on pool02, making it difficult to copy the data. It is also possible that the control panel of REMMINA, recognisable by the shadow at the top of the screen, disappears. If this happens, disconnect REMMINA from *spider* and restart the software.

A.2 Special weather events

As the KRT 3 is located outside on the roof of the Hautgebäude, it is subjected to the influence of the weather. In particular during and following special weather events, particular care has to be taken.

A.2.1 Strong winds and storms

The maximum wind speed during operation of the KRT 3 should not exceed 50 km h^{-1} (Beaufort scale ≤ 6). If the wind speed is higher, even if only in gusts, the telescope should be moved to the parking position to reduce the wind load. In the park position, the maximum permissible wind speed is 150 km h^{-1} .

A.2.2 Snowfall

Should it ever be necessary to clear the antenna of snow, the following procedure should be tried:

1. Move the antenna to the lowest possible position, i.e. to an elevation of 10° , to get the snow off the dish;
2. then change the azimuth angle until the dish is approximately perpendicular to the wind direction to blow the snow away;
3. point the telescope at the Sun to let the snow thaw. Alternatively, point it to the North–North–West (Azimuth 340°), so that the dish can be cleaned manually with a hand brush or something similar.

A.2.3 Extreme Temperatures

The permissible temperature range for operating the KRT 3 is from -10° to $+50^\circ$ Celsius. It should be taken into account that it is particularly hot on the roof in Sun shine in the summer. Therefore, the temperature in the shade should not exceed 40° Celsius.

A.2.4 Formation of ice

Should ice ever form on the dish, it should not affect operation as long as the ice formation is not too severe. If the thickness of the ice exceeds 5 mm, the weight of the ice is too high ($> 35 \text{ kg}$) and an attempt should be made to remove the ice. It may be sufficient to point the telescope towards the Sun. Otherwise, try to remove the ice carefully with a heater, fan or similar. Under no circumstances should the antenna be doused with hot water! Also, do not use alcohol-based solutions as is customary for car windscreens.

A.2.5 Hail

After heavy hailstorm, the structure of the antenna should be checked for possible deformation. If there is a large accumulation of hailstones in the dish, an attempt should first be made to remove them from the dish. This may be accomplished by lowering the dish to the lower

elevation limit of 10° . Otherwise, the hailstones must be removed manually with a hand brush or similar.

A.3 Hardware problems

The problems described below should only be attempted to be solved after consultation with the supervisor.

A.3.1 Loss of contact with the mount

It can happen that RADIOUNIVERSEPRO loses contact with the mount. This usually happens at the beginning of the observations directly when starting the software. This can be recognised by the fact that the mount is still oriented to the north and not to the west where the parking position is. If this is the case, the software should be terminated as described in Section 6 – of course without moving to the parking position, which is not possible. Then the receiver should first be switched off and then the connection to the mount. Then the connection to the mount is switched on again and then the receiver. Then the computer *spider* should be rebooted. This is a full system reboot, so to speak, which (hopefully) restores the functionality.

A.3.2 Loss of pointing accuracy

If the telescope has such a large pointing inaccuracy that it is difficult to locate the Sun, you can help yourself. Try to move the telescope carefully to the parking position. You can use the command `source=azel,<az>,<el>`, where *az* and *el* are the desired values for azimuth and elevation. The parking position is azimuth = 270° and elevation = 90° . Then the front of the mount should be parallel to the adapter plate on the west side. Of course, the telescope should also point vertically upwards. Then you have to change the values for azimuth and elevation until this is the case. Then the encoders of the mount can be reprogrammed.

A.3.3 Other issues

If the webcam does not work, it helps to restart it. Simply disconnect the power cable for a few seconds and reconnect it.

B Command line commands under LINUX

A ‘shell’ is a command interpreter that is structured as a command line with a prompt (‘prompt’). At the prompt, *often* displays the user name with/on which you are logged in, as well as the current working directory, e.g. `prak@pool02:~/Desktop$`. Where *prak* is the name of the user account, *pool02* is the name of the server, and `~/Desktop` is the directory you are currently in.

~ is always the abbreviation for your own 'home' directory, i.e. the directory where you can store your own data (in subdirectories if necessary). At the actual *prompt* '\$' can be used to enter commands and start programs. The prompt can also look different (in principle, each user can customize it), e.g., in some shells '>' is common in some shells.

ls (list files)

The ls command displays the files and directories in the current directory. The -l (long) option gives additional information such as access rights, user and group permissions, file size and modification date. The first 'd' in the output indicates that these are directories. These are available to the user prak and can be used by the members of the prak group.

Example: prak@pool02: ~> ls -l (press enter key)
drwxr-xr-x 2 prak prak 4096 Feb 25 13:50 downloads
-rw-r--r-- 1 prak prak 8980 Feb 27 2017 examples.desktop
drwxr-xr-x 16 prak prak 4096 Jun 17 16:46 krt

pwd (print working directory)

With pwd, the path to the current directory is displayed.

Example: prak@pool02:~/Desktop\$ ~> pwd
/prak/desktop

mkdir (make directory)

With mkdir you create a new directory.

Example: prak@pool02:~/Desktop\$ ~> mkdir radio .

cd (change directory)

With cd (directory name) you can change the current directory. change the current directory. Here '.' stands for the parent directory. cd without an argument is equivalent to cd ~ and will changes back to the home directory.

Examples: prak@pool02:~/desktop\$ ~> cd radio and cd ..

rmdir (remove directory)

With rmdir you remove a directory. This must be emptied beforehand.

Example: prak@pool02:~/Desktop\$ ~> rmdir radio

cp (copy)

You can copy files with cp. The first argument specifies the source file, which can be relative to the current working or absolute, e.g. to the home directory. the second argument specifies the destination. This can be either a directory or a new filename.

Example: prak@pool02:~/Desktop\$ ~> cp spectrum.py radio/spectrum_new.py

Explanation: Copy the file spectrum.py from the current directory to the subdirectory radio and rename it to spectrum_new.py.

mv (move)

Like cp, except that the file is moved. mv can also be used to rename the filename.

Example: `prak@pool02:~/Desktop$ ~> mv spectrum.py spectrum_new.py`

less (page by page display of text files)

With less a plain text file can be read in the window. With 'q' you can exit the display again.

Example: `prak@pool02:~/Desktop$ ~> less spectrum_new.py`

rm (remove)

With rm a file is deleted.

Example: `prak@pool02:~/Desktop$ ~> rm spectrum_new.py`

man (manual)

With man command name, a help text with all options for a specific command is output.

Example: `prak@pool02:~/Desktop$ ~> man rm`

C Minieinführung in PYTHON

C.1 Grundlagen

PYTHON ist eine kommandozeilenbasierte, frei verfügbare Programmiersprache, die zum Plotten von Funktionen und Daten sehr geeignet ist. Außerdem lassen sich beliebige Funktionen an Daten fitten, um Parameter physikalischer Modelle an Messreihen anzupassen und den Fehler beim Fitten zu bestimmen. Diese Zusammenfassung soll einen Überblick über die gebräuchlichsten PYTHON-Kommandos geben. Es wird empfohlen sich eine aktuelle Installation von ANACONDA¹⁹ zuzulegen, da diese schon alle nötigen Pakete, vor allem ASTROPY (version 4.0 oder höher) enthält. Hat man ANACONDA installiert, startet man den *Anaconda-Navigator*. Damit lässt sich dann ein JUPYTER-Notebook öffnen, mit dem man im Webbrowser PYTHON-scripte ausführen kann. Alternativ lässt sich auch ein *Terminal* öffnen. Dazu im Anaconda-Navigator auf *Environments* klicken, und dann auf das 'Play' Icon neben *base (root)* klicken. Damit hat man nun Zugriff auf eine Shell mit PYTHON. Im folgenden nehmen wir an, dass wir diese Shell benutzen.

C.2 Reading files

With PYTHON, data and functions can be displayed in many different ways. In the following, we will show how a 2D diagram with a linear fit can be displayed. To do this, we create a PYTHON script, which we can call for example `fit.py`. The command

¹⁹<https://www.anaconda.com/products/individual>

Table 4: Preamble of PYTHON scripts

Command	Description
<code>import sys</code>	System-specific functions
<code>import numpy</code>	Scientific Package
<code>import matplotlib.pyplot as plt</code>	Plotting data
<code>from scipy.optimize import curve_fit</code>	Least-squares fit

```
$ python fit.py book.txt
```

then applies the script to the file `book.txt`. As always, the ‘\$’ is not entered, but is just the command line prompt.

In the example, you have to pass the name of the file to be read in. This can be done with `filename = sys.argv[1]`. Here we call the function `sys` of the system. To do this, we include the command `import sys` in the preamble of the PYTHON script. Other commands needed for the preamble are shown in table 4. We assume that the file to be read has two columns, with x and y values. So we now read the file:

```
x=[]
y=[]
sigma_y = []
filename = sys.argv[1]
with open(filename) as inf:
    for line in inf:
        if not line.startswith('#'):
            parts= line.split()
            x.append(float(parts[0]))
            y.append(float(parts[1]))
            sigma_y.append(float(parts[2]))
```

Here, attention must be paid to the correct **insertion** of the lines. We have assumed that comment lines in the file to be read in begin with a so-called hash mark ‘#’. Now the result that e.g. the x -values were read in correctly can be checked with a simple `print('x=', x)` if you want to check the result. We have also given an estimated error for the y values, which we read in as σ_y from the third column...

C.3 Plotting data

PYTHON can directly plot the lists of x and y values stored as data in a file. The different columns can be separated by, for example, one space, several spaces or a tabulator. The first column indicates the x values and the second the y values. The assignment of the columns can be modified with the command `parts[2]`, e.g. to the third column. It is also possible to

perform arithmetic operations on a column. For example, if you want to divide all y -values by two, you can do this with:

```
plt.plot(x, 0.5*np.asarray(y), marker='o', color='r')
plt.title('Unser Plot')
plt.xlabel('x-Werte (einheiten)')
plt.ylabel('y-Werte (einheiten)')
plt.show()
```

Note that we have converted the y values into an *Numpy Array* to perform the multiplication. In general, you can also define a function:

```
def f(x):
    return x**2
```

This defines a quadratic function $f(x) = x^2$. In the example above we add:

```
plt.plot(x, f(np.asarray(x)), marker='+', color='b')
```

to display this function with the corresponding y values $y = f(x)$. Further options for the *matplotlib* plot command are given in table 5.

Table 5: Das *matplotlib* plot-Kommando

Command	Description	Example
plt.plot	Plots the corresponding file(s), function(s)	plt.plot(x, y)
...	Plots the x values as a function of the y values	plt.plot(y, x)
plt.errorbar	Shows data with errorbars	plt.errorbar(x, y, sigma_y)
...	Does not draw lines between data points	plt.errorbar(..., fmt = 'o')
...	Displays the data points as dots instead of crosses	plt.errorbar(..., marker = '+')
...linewidth	Curve with given width	plt.errorbar(..., linewidth=2)
...markersize	Data points with a given size	plt.errorbar(..., markersize=5)
...capsize	Error bars with end caps	plt.errorbar(..., capsize = 2.5)

C.4 Scaling, labelling, and output format

A logarithmic representation of the y -axis is given by the command:

```
plt.yscale('log')
```

Analogously, the x -axis can also be displayed logarithmically. Often it is necessary to visualise only a partial area of a function or a data set. For this purpose one can use:

```
plt.xlim(xmin, xmax)
```

to display only the points with x -values within the interval `[xmin:xmax]`. The same also works for the y -axis.

To create Greek characters or superscript or subscript characters, you can use the $\text{\LaTeX}$ standard (see Table 6).

Table 6: Labels with greek letters

Symbol	plt Syntax	Symbol	plt Syntax
a^b	<code>plt.xlabel(r'\$a^b\$')</code>	a_b	<code>plt.xlabel(r'\$a_b\$')</code>
α	<code>plt.xlabel(r'\$\alpha\$')</code>	β	<code>plt.xlabel(r'\$\beta\$')</code>
γ	<code>plt.xlabel(r'\$\gamma\$')</code>	ϑ	<code>plt.xlabel(r'\$\vartheta\$')</code>

To label the y -axis of the above parabola with $\pi(x)$, use the following :

```
plt.xlabel(r'$\pi(x)$')
```

In general, the scales and labels can be specified with attributes, some of which are shown in Table 7. The plots can be exported lossless to png files.

Table 7: The attributes for the *matplotlib* plot command

Attribute	Description	Example
<code>plt.title</code>	Sets the title caption	<code>plt.title('Unser Plot')</code>
<code>plt.xlabel</code>	Sets the x -axis labelling	<code>plt.xlabel('x-Werte')</code>
<code>plt.legend</code>	Legend for the plot	<code>plt.legend(loc='upper right')</code>
<code>plt.xlim</code>	Specify definition range	<code>plt.xlim(-1, 1)</code>
<code>plt.yscale</code>	Logarithmic representation of an axis	<code>plt.yscale('log')</code>
<code>plt.grid</code>	Plot grid	<code>plt.grid(True)</code>

C.5 Fitting a function

With PYTHON you can also fit a function with the least squares method. We apply this to our book example and define a linear function to fit the thickness as a function of the number of pages with $f(x) = ax + b$:

Table 8: Thickness of a book as a function of the number of pages.

Pages	Thickness (cm)
100	0.7
200	1.0
300	1.3
400	1.7
500	2.0
600	2.5

```
def f(x):
    a*x+b
```

Now we need initial values for the slope a and the y -axis section b , so that a solution can be found numerically.:

```
initial = np.array([1., 1.])
```

Then the best fit can be found like this:

```
popt, pcov = curve_fit(f, np.asarray(x), np.asarray(y), initial,
np.asarray(sigma_y))
```

The results for slope a and y -axis section b can be output as follows:

```
print('a =', pop[0], '+/-', pcov[0,0]**0.5, 'cm')
print('b =', pop[1], '+/-', pcov[1,1]**0.5, 'cm')
```

Now we would like to calculate the recycled χ^2 :

```
model = f(np.asarray(x),popt[0],popt[1])
r = np.asarray(y) - model
chisq = np.sum((r/np.asarray(sigma_y))**2)
df = len(y) - 2.0
print('Reduced chisq =', chisq/df)
print('Degrees of freedom =', df)
```

Here we have assumed that we have two fit parameters (a, b). Thus the number of degrees of freedom df (degrees of freedom) is equal to the number of data points minus the number of fit parameters.

C.6 Quick start

In the following, the generation of a plot and its output as png-file will be demonstrated by way of example. Let us assume that we have measured the thickness of a book with the help of a ruler. We measure the thickness as a function of the number of pages (exactly: pages) including one cover. The result is shown in Table 8. Copy the PYTHON script `fit.py` and the data from

the book measurement `book.txt` from the Desktop directory to your working directory.²⁰ You can use the PYTHON script as a template for creating your own plots. It is a good idea to create a separate PYTHON script for each plot so that you can recreate the plots at any time. In the PYTHON script `fit.py`, lines preceded by '#' are comment lines. These and the empty lines are ignored by PYTHON, so you can document your script with comments.

Detailed help can be found in the **matplotlib** documentation at:

<https://matplotlib.org>

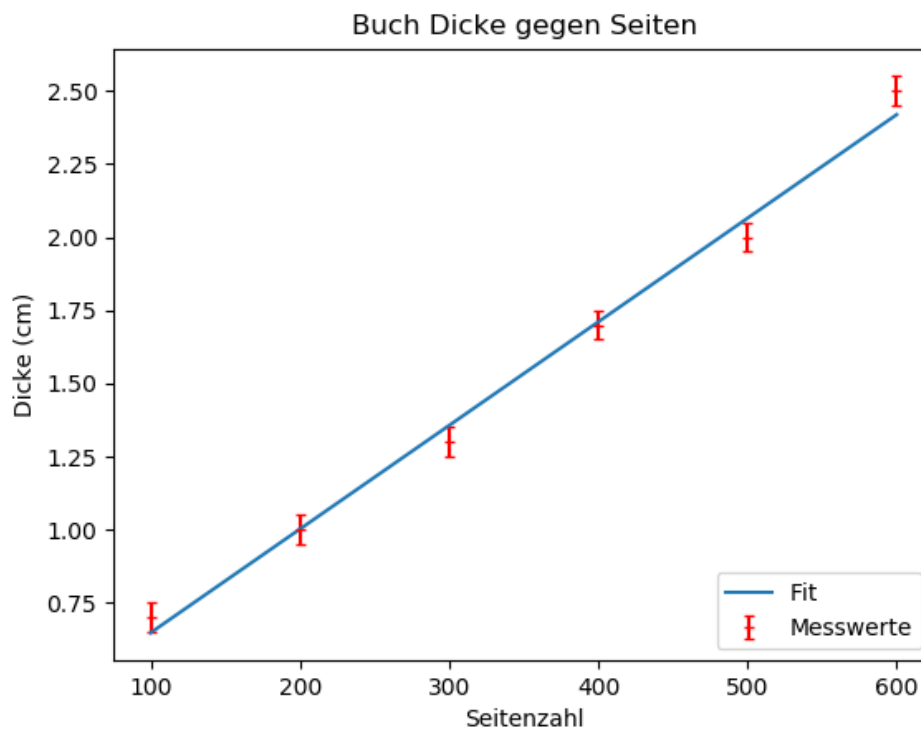


Figure 20: Plot created with PYTHON. The following script was used `fit.py`

What is in the script `fit.py` has already been explained, including the fitting of the straight line. In order to show the straight line in an attractive way, we define a few additional x values and the corresponding function values:

```
np.arange(np.amin(x), np.amax(x), np.amax(x)-np.amin(x))/100.  
model_plot = f(x_model, popt[0], popt[1])
```

Now we plot all this with *matplotlib*:

```
plt.errorbar(x, y, sigma_y, marker='_', color='r', fmt='o',  
            markersize = 5, capsize = 2.5, label='Messwerte')  
plt.plot(x_model, model_plot, label='Fit')
```

²⁰ Alternatively, the files can also be found on the observatory's *Nextcloud*

```
plt.legend(loc='lower right')
plt.title('Buch Dicke gegen Seiten')
plt.xlabel('Seitenzahl')
plt.ylabel('Dicke (cm)')
plt.show()
```

The plot shown should correspond to the one in Fig. 20. Now the plot can be easily exported as a png file. Simply click on the 'disk'. If you wish to use a different graphic format for the output, choose from the list of all available output formats. Furthermore, the script should give the following text output in the shell:

```
=====Best-fitting results=====
a = 0.0035428571406836973 +/- 0.0001547435970236191 cm
b = 0.293333333333183564 +/- 0.06026397154829759 cm
Reduced chisq = 1.6761904761904745
Degrees of freedom = 4.0
=====
```

Since the value of the reduced χ^2 is in the range 1–2, we can see that the estimated error is approximately correct.



Figure 21: Old KRT 3 in March 2018.

D Short history of the KRT 3

The KRT 3 is a radio telescope that has been in operation at Hamburg Observatory since 2011 (now in its second incarnation). The radio telescope has a parabolic mirror with a diameter of 3 m and a digital receiver for wavelengths of 21 cm (1420 MHz). With the KRT 3, the continuum radiation of bright radio sources and the 21 cm line of atomic hydrogen can be observed. Here is a brief history of the KRT 3 at Hamburg observatory:

- 2010: proposal by Dieter Engels that the observatory procures a radio telescope for a possible lab course experiment.
- 2011: acquisition of a 3.1-m radio telescope from Custom Astronomical Support Services, Inc. in the USA, based on the *Small Radio Telescope (SRT)* of the MIT Haystack Observatory²¹ (see Fig. 21).
- March 2016: first instructions for the advanced lab-course experiment ‘Radio Astronomy’ written by Dieter Engels. The suggested data analysis was based on table calculation programs (e.g. EXCEL, OPENOFFICE) and the plotting program GNUPLOT.
- March 2016: First practical experiment with the old KRT 3 supervised by Dieter Engels.
- February 2017–May 2019: advanced lab-course experiment with the old KRT 3 supervised by Volker Heesen. Transition to PYTHON to replace the table calculation programs for the analysis of the H I spectra.
- 26 March 2019: Last observation with the old KRT 3.
- April 2019: Disassembly of the old KRT 3.
- July 2019: Installation of the new KRT 3, a SPIDER 300A model by PrimaluceLab SpA based in Italy, unofficially also called ‘Dieter Engels Telescope’ (see Fig. 22).²²
- August 2019: Installation of RADIOUNIVERSEPRO version 1.4.5.
- August 14, 2019: ‘First Light’ with the new KRT 3.
- August 2019: First lab course experiment with the new KRT 3.
- October 2020: Data analysis for the lab course experiment now fully transitioned to PYTHON.
- February: Installation of the noise generator.
- June 2021: Installation of RADIOUNIVERSEPRO version 1.6.7.
- March 2022: Noise generator becomes functional.

²¹<https://www.haystack.mit.edu/haystack-public-outreach/srt-the-small-radio-telescope-for-education/>

²²<https://www.radio2space.com/product/spider-300a-advanced-radio-telescope/>



Figure 22: New KRT 3-Radiotelescope of Hamburg Sternwarte.